MAT 830 – Topics in Algebra: The stable category of a commutative Gorenstein ring

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Preface

These are notes from a graduate topics course I am currently running during the Fall 2024 semester at Syracuse University. The main object of study is the stable (or singularity) of a Gorenstein ring; that being said, a major theme of the notes will be "duality" over such rings.

Chapter 1

Background

1.1 Notation

Throughout the text R will be a commutative ring. We let $\mathsf{Mod}(R)$ denote the abelian category of R-modules. We will also be concerned with several of its (full) subcategories:

$$\begin{aligned}\mathsf{Flat}(R) &:= \text{all flat } R\text{-modules}\\ \mathsf{Inj}(R) &:= \text{all flat } R\text{-modules}\\ \mathsf{Proj}(R) &:= \text{all projective } R\text{-modules}\\ \mathsf{art}(R) &:= \text{all artinian } R\text{-modules}\\ \mathsf{noeth}(R) &:= \text{all noetherian } R\text{-modules}\,.\end{aligned}$$

f.g. = lowercase

When R is noetherian, $\mathsf{noeth}(R)$ consists of exactly the finitely generated R-modules; in this case, we use the more standard notation $\mathsf{mod}(R)$ for this category. Also, in this setting, write $\mathsf{proj}(R)$ for the category of finitely generated projective R-modules; that is to say,

$$\mathsf{proj}(R) := \mathsf{mod}(R) \cap \mathsf{Proj}(R)\,.$$

Finally, we write $\mathsf{fl}(R)$ for the finite length R-modules. That is,

$$\mathsf{fl}(R) := \mathsf{mod}(R) \cap \mathsf{art}(R)\,.$$

Noeth. + Art. $\Leftrightarrow$ fl

1.2 Matlis duality

In many regards, Matlis duality is a consequence of understanding the duality theory in the next proposition in a particular instance (namely, when one dualizes with respect to the injective hull of the residue field of a local ring; cf. Lemma 1.2.10).

Proposition 1.2.1. *Fix an injective R-module E. The endofunctor $(-)^\vee := \mathrm{Hom}_R(-,E)$ on $\mathsf{Mod}(R)$, restricts to functors $\mathsf{Flat}(R) \rightleftarrows \mathsf{Inj}(R)$. That is to say,*

(a) *If F is flat, then $F^\vee$ is injective;*

(b) *if I is injective, then $I^\vee$ is flat.*

Remark 1.2.2. Proposition 1.2.1(a) is an immediate consequence of adjunction. Part (b) is consequence of the following (more general) lemma.

(a) • E inj. $\Leftrightarrow$ $\mathrm{Hom}_R(-,E)$ exact since def. of E is "extending the map": $A \hookrightarrow B$ "into E".

• By adjunction, $\mathrm{Hom}_R(A, \mathrm{Hom}_R(F,E)) \cong \mathrm{Hom}_R(A \otimes F, E)$ thus $\mathrm{Hom}_R(-, F^\vee)$ exact.

(F exact, $\otimes$ exact)

② Def: M fin. presented: $R^p \to R^q \to M \to 0$.

Lemma 1.2.3. *For R-modules M, N, E, there is a natural map $M \otimes_R \mathrm{Hom}_R(N,E) \to \mathrm{Hom}_R(\mathrm{Hom}_R(M,N),E)$ given by evaluation:*

$$m \otimes \alpha \mapsto [\beta \mapsto \alpha(\beta(m))]\,.$$

When M is finitely presented and E is injective this map is an isomorphism.

Proof. It is easy to check that the map is natural in each of its components and an isomorphism for any choice of N, E whenever M is a finite rank free R-module. Now fix a finite free presentation $F \to G \to M \to 0$, and consider the commutative diagram

$$\begin{array}{ccccccc} F \otimes \mathrm{Hom}_R(N,E) & \longrightarrow & G \otimes \mathrm{Hom}_R(N,E) & \longrightarrow & M \otimes \mathrm{Hom}_R(N,E) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ \mathrm{Hom}_R(\mathrm{Hom}_R(F,N),E) & \longrightarrow & \mathrm{Hom}_R(\mathrm{Hom}_R(G,N),E) & \longrightarrow & \mathrm{Hom}_R(\mathrm{Hom}_R(M,N),E) & \longrightarrow & 0\,. \end{array}$$

The top row is exact as $- \otimes_R \mathrm{Hom}_R(N,E)$ is always right exact, while the bottom row is exact since E is injective. Finally, it remains to note that left two vertical maps are isomorphisms since F, G are finite rank free, and so it follows that the third vertical is also an isomorphism, completing the proof of the lemma. □

Proof of Proposition 1.2.1(b). The essential point is that Lemma 1.2.3 implies the isomorphism below holds

$$- \otimes_R I^\vee = - \otimes_R \mathrm{Hom}_R(I,E) \cong \mathrm{Hom}_R(\mathrm{Hom}_R(-,I),E)$$

on the category of the finitely presented R-modules. As the right-hand side is exact, it follows that

$$\mathrm{Tor}_1^R(-, I^\vee) = 0 \quad \text{on} \quad \mathsf{mod}(R)\,.$$

Also, since every R-module is a filtered colimit of finitely generated R-modules, and filtered colimits commute with Tor, it follows that $\mathrm{Tor}_1^R(-, I^\vee)$ vanishes on all of $\mathsf{Mod}(R)$. Hence, $I^\vee$ is flat as claimed. □

For the remainder of the section, assume that $(R, \mathfrak{m}, k)$ is noetherian local. We set $E := E_R(k)$; here for an R-module M, we write $E_R(M)$ for its injective hull. Namely, $E_R(M)$ is the maximal essential extension of R containing M; equivalently, $E_R(M)$ is the minimal injective R-module containing M.

Definition 1.2.4. The **Matlis dual** functor is the endofunctor $(-)^\vee := \mathrm{Hom}_R(-,E)$ on $\mathsf{Mod}(R)$.

Lemma 1.2.5. *E is a cogenerator for $\mathsf{Mod}(R)$. That is to say, an R-module M is zero if and only if $M^\vee$ is zero.*

Proof. If $M \neq 0$, there exists an ideal $I \subsetneq R$ such $R/I \subseteq M$. Furthermore, as the natural map $R/I \to k \to E$ defines a nonzero elements of $R/I^\vee$, and since $M^\vee \to (R/I)^\vee$ is surjective, it follows that $M^\vee \neq 0$. □

Remark 1.2.6. If $\varphi\colon R \to (S, \mathfrak{n}, \ell)$ is a finite local homomorphism (that is to say, $\varphi(\mathfrak{m}) \subseteq \mathfrak{n}$ and S is module finite over R), then $\mathrm{Hom}_R(S,E) \cong E_S(\ell)$. In particular, Matlis duality over R and S are the same on the category of S-modules.[1]

We will apply this when S is a quotient of R in the sequel. Furthermore, this fact can also be used to calculate E. Namely, it will be shown (see Lemma 1.2.10) that $E \cong E_{\widehat{R}}(k)$ where $\widehat{R}$ is $\mathfrak{m}$-adic completion of R. Furthermore, by the Cohen structure theorem we can write $\widehat{R} = Q/I$. Namely, it suffices to compute $E_Q(k)$ when Q is a regular local ring with residue field k to understand E.

We now state the main result of the section.

Theorem 1.2.7 (Matlis duality). *Assume $(R, \mathfrak{m}, k)$ is a complete noetherian local ring and set $E := E_R(k)$. The Matlis dual functor $(-)^\vee = \mathrm{Hom}_R(-,E)$ restricts to equivalences, with inverse being itself, of the horizontal functors below*

[1] This is part of Exercise 1 on Discussion Problems 1.

$$\begin{array}{ccc} \mathsf{mod}(R) & \xrightarrow[\equiv]{(-)^\vee} & \mathsf{art}(R) \\ \subseteq\uparrow & & \uparrow\subseteq \\ \mathsf{fl}(R) & \xrightarrow[\equiv]{(-)^\vee} & \mathsf{fl}(R)\,. \end{array}$$

In particular, the following hold:

(a) *E is artinian and $E^\vee = R$;*

(b) *$M \xrightarrow{\cong} M^{\vee\vee}$ is an isomorphism whenever M is artinian or finitely generated;*

(c) *M is finitely generated if and only if $M^\vee$ is artinian.*

Remark 1.2.8. Returning to the more general setting that R is a noetherian local ring, by Proposition 1.2.1 the R-module $E^\vee$ is flat. Identifying the flat R-module $E^\vee$ is the essential part of Matlis duality. Indeed, for instance, by Lemma 1.2.3 it follows that for a finitely generated M we have that the biduality map is the canonical one

$$M \to M^{\vee\vee} \cong M \otimes_R \operatorname{Hom}_R(E,E) = M \otimes_R E^\vee\,;$$

therefore, it would suffice to show $E^\vee \cong \widehat{R}$ to deduce the map is an isomorphism when R is complete.

Before giving the proof of Theorem 1.2.7, we need some preliminary results.

Lemma 1.2.9. *If N is finite length, then* $\operatorname{length}_R(N) = \operatorname{length}_R(N^\vee)$ *and $N \xrightarrow{\cong} N^{\vee\vee}$. In particular, if R is artinian, then $R^{\vee\vee} = \operatorname{Hom}_R(E,E) \cong R$.*

Proof. This is clear for k since $k^\vee = k$, and now a simple induction on length yields the result. □

Lemma 1.2.10. (a) *E is naturally an $\widehat{R}$-module and $E \cong E_{\widehat{R}}(k)$;*

(b) $\widehat{R} \cong \operatorname{End}_R(E)$.

Proof. The fact E is an $\widehat{R}$-module follows directly from the fact that it is $\mathfrak{m}$-power torsion. Also, as E is $\mathfrak{m}$-power torsion we have the equality below

$$E = \bigcup_{i\geqslant 0}(0 :_E R/\mathfrak{m}^i) \cong \operatorname{colim} \operatorname{Hom}_R(R/\mathfrak{m}^i, E) \cong \operatorname{colim} E_{R/\mathfrak{m}^i}(k) \tag{1.2.10.1}$$

where the first isomorphism is the standard one and the second one is from Remark 1.2.6. As $R/\mathfrak{m}^i \to \widehat{R}/\mathfrak{m}^i\widehat{R}$ is an isomorphism for each i, then by (1.2.10.1) it follows that the natural map $E \to E_{\widehat{R}}(k)$ is an isomorphism, which completes the proof of part (a).

For part (b), consider the isomorphisms

$$\begin{aligned} \operatorname{End}_R(E) &= \operatorname{Hom}_R(E,E) \\ &\cong \operatorname{Hom}_R(\operatorname{colim} E_{R/\mathfrak{m}^i}(k), E) \\ &\cong \lim \operatorname{Hom}_R(E_{R/\mathfrak{m}^i}(k), E) \\ &\cong \lim \operatorname{Hom}_{R/\mathfrak{m}^i}(E_{R/\mathfrak{m}^i}(k), \operatorname{Hom}_R(R/\mathfrak{m}^i, E)) \\ &\cong \lim \operatorname{Hom}_{R/\mathfrak{m}^i}(E_{R/\mathfrak{m}^i}(k), E_{R/\mathfrak{m}^i}(k)) \\ &\cong \lim R/\mathfrak{m}^i \\ &\cong \widehat{R}\,; \end{aligned}$$

the first isomorphism uses (1.2.10.1), the fourth uses Remark 1.2.6, and the fifth uses Lemma 1.2.9, while the others are obvious isomorphisms. □

Remark 1.2.11. Recall that an R-module is artinian if and only if it is a submodule of $E^{\oplus n}$ for $n > 0$;[2] The smallest such n can be taken to the k-vector space dimension of $(0 :_M \mathfrak{m})$, the socle of M.

We are now to ready prove the theorem on Matlis duality.

Proof of Theorem 1.2.7. For (a), it remains to show E is artinian as $E^\vee \cong R$ was justified in Lemma 1.2.10. Consider a descending chain of submodules

$$E = E_0 \supseteq E_1 \supseteq E_2 \supseteq \dots ,$$

then by applying $(-)^\vee$ we get surjections

$$R \to E_1^\vee \to E_2^\vee \to \dots .$$

This defines an ascending chain of ideals in R which must stabilizies as R is noetherian. Therefore, $E_i^\vee = E_{i+1}^\vee$ for all $i \gg 0$, and from Lemma 1.2.5 it follows that $E_i = E_{i+1}$ for all $i \gg 0$, as needed.

For (b), the desired isomorphism holds for finitely generated modules by Remark 1.2.8 and Lemma 1.2.10. Note that by Lemma 1.2.10, we have

$$E^{\vee\vee} = R^\vee = E . \tag{1.2.11.1}$$

. Now if M is artinian, then we fix an injection $M \to E^{\oplus n}$; cf. Remark 1.2.11. Also, as E is artinian from part (a), we can again apply Remark 1.2.11 to obtain a left exact sequence

$$0 \to M \to E^{\oplus n} \to E^{\oplus m} .$$

Applying $(-)^{\vee\vee}$ gives the following commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & E & \longrightarrow & E \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M^{\vee\vee} & \longrightarrow & E^{\vee\vee} & \longrightarrow & E^{\vee\vee} ; \end{array}$$

it remains to note from (1.2.11.1) that $M \xrightarrow{\cong} M^{\vee\vee}$ when M is artinian.

For (c), if M is finitely generated, then the surjection $R^{\oplus n} \to M$ dualizes to an injection $M^\vee \to E^{\oplus n}$. Now by (b), since E is artinian, it follows that $M^\vee$ is as well. Next, if M is artinian, the injection $M \to E^{\oplus n}$ dualizes to the surjection $R^n \to M^\vee$ and so it follows that $M^\vee$ is noetherian. □

1.3 Derived category

First, we recall some notation and terminology, and set some conventions regarding chain complexes of R-modules. Much of the section is sensible for arbitrary rings, where one needs to declare whether they are working with left or right modules; however, we leave the necessary adjustment to the interested reader.

1.3.1. Recall that an *R-complex* is a graded R-module $X = \{X_i\}_{i\in\mathbb{Z}}$ equipped with a degree -1 map ∂^X (that is to say, $\partial^X(X_i) \subseteq X_{i-1}$ for all $i \in \mathbb{Z}$) such that $\partial^X \circ \partial^X = 0$. Namely, it is a sequence of R-linear maps

$$\dots \to X_{i+1} \xrightarrow{\partial^X_{i+1}} X_i \xrightarrow{\partial^X_i} X_{i-1} \to \dots$$

where $\partial^X_i = \partial^X|_{X_i}$ for each $i \in \mathbb{Z}$. An element x of X belongs to a unique module X_i; we say the degree of x is i, and write $|x| = i$.

Given an R-complex X, its *cycles* and *boundaries* are

$$\mathrm{Z}(X) := \ker \partial^X \quad \text{and} \quad \mathrm{B}(X) := \partial^X(X) ,$$

[2] This is part Exercise 1 of Discussion Problems 1.

respectively. It is clear that $\mathrm{B}(X)$ is a graded submodule of $\mathrm{Z}(X)$; the *homology* of X is

$$\mathrm{H}(X) := \frac{\mathrm{Z}(X)}{\mathrm{B}(X)}.$$

Explicitly, $\mathrm{H}_i(X) = \dfrac{\ker \partial^X_i}{\partial^X(X_{i+1})}$ is the ith *homology module* of X. For a cycle z, we write its homology class in $\mathrm{H}(X)$ as $[z]$. We say X is *acyclic* if $\mathrm{H}(X) = 0$.

A *morphism of complexes* is a map of graded R-modules $\alpha\colon X \to Y$ such that α commutes with the differentials. More precisely, α is a sequence of maps $\alpha = \{\alpha_i\colon X_i \to Y_i\}_{i\in\mathbb{Z}}$ where each α_i is an R-linear map and the following commute

$$\begin{array}{ccc} X_i & \xrightarrow{\partial^X_i} & X_{i-1} \\ {\scriptstyle\alpha_i}\downarrow & & \downarrow{\scriptstyle\alpha_{i-1}} \\ Y_i & \xrightarrow[\partial^Y_i]{} & Y_{i-1} \end{array} .$$

A morphism of complexes $\alpha\colon X \to Y$ induces a map of graded R-modules on homology

$$\mathrm{H}(\alpha)\colon \mathrm{H}(X) \to \mathrm{H}(Y) \quad \text{given by} \quad [z] \mapsto [\alpha(z)]\,;$$

the morphism α is called a *quasi-isomorphism* if the induced map $\mathrm{H}(\alpha)$ is an isomorphism, and in this case we decorate it as $\alpha\colon X \xrightarrow{\simeq} Y$.

Notation 1.3.2. Let $\mathsf{C}(R)$ denote the category of complexes of R-modules. That is to say, $\mathsf{C}(R)$ is the category whose objects are R-complexes and morphisms are maps of complexes; it is to see that $\mathsf{C}(R)$ is an abelian category.

1.3.3. Given an R-complex X we define ΣX in $\mathsf{C}(R)$ by $(\Sigma X)_i = X_{i-1}$ and $\partial^{\Sigma X}_i = -\partial^X_{i-1}$ for each $i \in \mathbb{Z}$. This defines a an auto-equivalence of $\mathsf{C}(R)$, and let Σ^{-1} denote its inverse; also let Σ^n denote the appropriate n-fold composition.

The *mapping cone* of a map of R-complexes $\alpha\colon X \to Y$ is the R-complex $\operatorname{cone}\alpha$ with underlying graded R-module $Y \oplus \Sigma X$ and differential

$$\partial^{\operatorname{cone}\alpha} = \begin{pmatrix} \partial^Y & \alpha \\ 0 & \partial^{\Sigma X} \end{pmatrix} = \begin{pmatrix} \partial^Y & \alpha \\ 0 & -\partial^X \end{pmatrix}.$$

Explicitly, $(\operatorname{cone}\alpha)_i = Y_i \oplus X_{i-1}$ and

$$\partial^{\operatorname{cone}\alpha}(y,x) = (\partial^Y(y) + \alpha(x), -\partial^X(x))\,.$$

There is an exact sequence of R-complexes

$$0 \to Y \to \operatorname{cone}\alpha \to \Sigma X \to 0\,,$$

with the maps being the obvious ones; this exact sequence induces an exact sequence in homology

$$\mathrm{H}(X) \xrightarrow{\mathrm{H}(\alpha)} \mathrm{H}(Y) \to \mathrm{H}(\operatorname{cone}\alpha) \to \mathrm{H}(\Sigma X) \xrightarrow{\mathrm{H}(\Sigma\alpha)} \mathrm{H}(\Sigma Y)\,.$$

That is to say, there is an exact sequence of R-modules

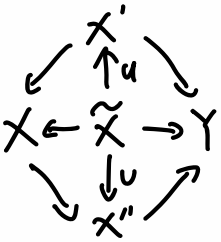
$$\dots \to \mathrm{H}_i(X) \xrightarrow{\mathrm{H}_i(\alpha)} \mathrm{H}_i(Y) \to \mathrm{H}_i(\operatorname{cone}\alpha) \to \mathrm{H}_{i-1}(X) \xrightarrow{\mathrm{H}_{i-1}(\alpha)} \mathrm{H}_{i-1}(Y) \to \dots\,.$$

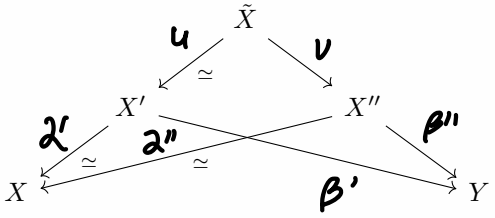
It follows that $\operatorname{cone}\alpha$ is acyclic if and only if α is a quasi-isomorphism.

(2) Localisation of C(R) w.r.t. quasi-isom.

Definition 1.3.4. The **derived category** of R, denoted $\mathsf{D}(R)$, is obtained by formally inverting all quasi-isomorphisms. That is, $\mathsf{D}(R)$ solves the following universal problem: there is a functor $q\colon \mathsf{C}(R) \to \mathsf{D}(R)$ such that $Q(\alpha)$ is an isomorphism in $\mathsf{D}(R)$ for every quasi-isomorphism of R-complexes α, and any functor $q'\colon \mathsf{C}(R) \to \mathsf{D}'$ that inverts all quasi-isomorphisms uniquely factors through q. We write isomorphisms in $\mathsf{D}(R)$ with $\simeq$.

Construction 1.3.5. The objects of $\mathsf{D}(R)$ are R-complexes. Morphisms are given by a *calculus of fractions*. Namely, a morphism $X \to Y$ in $\mathsf{D}(R)$ corresponds to a diagram of R-complexes diagrams $X \xleftarrow{\simeq} X' \to Y$ up to the equivalence relation given by $X \xleftarrow{\simeq} X' \to Y$ is equivalent to $X \xleftarrow{\simeq} X'' \to Y$ if these fit into the following diagram of R-complexes that commute *up to homotopy*: means $\alpha' u \simeq_h \alpha'' v$, $\beta' u \simeq_h \beta'' v$.





homot. equiv.

X̃ is a common refinement.

Given a diagram $X \xleftarrow[\beta]{\simeq} X' \xrightarrow[\alpha]{} Y$, we write $\frac{\alpha}{\beta}$ for its equivalence class. For composition, given $X \xleftarrow[\beta]{\simeq} X' \xrightarrow[\alpha]{} Y$ and $Y \xleftarrow[\delta]{\simeq} Y' \xrightarrow[\gamma]{} Z$ we define the composition $\frac{\delta}{\gamma}\frac{\alpha}{\beta} := \frac{\gamma\alpha'}{\delta'\beta}$ where the following commutes up to homotopy

Here X'' is P ≃ X'.

$$\begin{array}{ccc} X' & \xrightarrow{\alpha} & Y \\ {\scriptstyle\delta'}\uparrow{\scriptstyle\simeq} & & {\scriptstyle\simeq}\uparrow{\scriptstyle\delta} \\ X'' & \xrightarrow{\alpha'} & Y' \end{array}.$$

Remark 1.3.6. The derived category $\mathsf{D}(R)$ is a triangulated category with Σ being the suspension functor and exact triangles isomorphic to

$$X \xrightarrow{\alpha} Y \to \operatorname{cone}\alpha \to$$

where α is a morphism of R-complexes. In particular, given an exact triangle

$$X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \to$$

in $\mathsf{D}(R)$, there is a long exact sequence in homology

$$\dots \to \mathrm{H}_i(X) \xrightarrow{\mathrm{H}_i(\alpha)} \mathrm{H}_i(Y) \xrightarrow{\mathrm{H}_i(\beta)} \mathrm{H}_i(Z) \to \mathrm{H}_{i-1}(X) \xrightarrow{\mathrm{H}_{i-1}(\alpha)} \mathrm{H}_{i-1}(Y) \to \dots;$$

X ≅ Y ⇑ so that cone(α) = 0 for α quasi-isom.

cf. 1.3.3. One way to to see all this is that $\mathsf{D}(R)$ is equivalent to Verdier quotient of the homotopy category $\mathsf{K}(R)$ by the category of acyclic R-complexes $\mathsf{K}_{\mathrm{ac}}(R)$; that is to say $\mathsf{D}(R) \equiv \mathsf{K}(R)/\mathsf{K}_{\mathrm{ac}}(R)$. As Verdier quotients inherits a triangulated category structure from the larger category, this prescribes $\mathsf{D}(R)$ with such a structure which is the one recalled above. Later we will show $\mathsf{K}(R)$ is triangulated and describe this in a bit more detail.

morphisms up to equiv.

(3)

Definition 1.3.7. Fix R-complexes X, Y. We define their **Hom complex** $\operatorname{Hom}_R(X,Y)$ and **tensor complex** $X \otimes_R Y$ as:

(a) $\operatorname{Hom}_R(X,Y)_i := \prod_{j\in\mathbb{Z}} \operatorname{Hom}_R(X_j, Y_{i+j})$ and $\partial^{\operatorname{Hom}_R(X,Y)}(\alpha) := \partial^Y\alpha - (-1)^{|\alpha|}\alpha\partial^X$.

(b) $(X \otimes_R Y)_i := \bigoplus_{j\in\mathbb{Z}} X_j \otimes_R Y_{i-j}$ and $\partial^{X\otimes_R Y}(x\otimes y) := \partial^X(x)\otimes y + (-1)^{|x|}x\otimes\partial^Y(y)$.

These define bifunctors on $\mathsf{C}(R)$ in the obvious way, and we will want to deal with their derived versions which are bifunctors on $\mathsf{D}(R)$. First we define the classes of complexes one resolves a complex with to compute these functors.

i) **Definition 1.3.8.** (a) An R-complex P is **semiprojective** if $\mathrm{Hom}_R(P,-)$ is exact and preserves quasi-isomorphisms.

(b) An R-complex I is **semiinjective** if $\mathrm{Hom}_R(-,I)$ is exact and preserves quasi-isomorphisms.

(c) An R-complex F is **semiflat** if $F\otimes_R -$ is exact and preserves quasi-isomorphisms.

Example 1.3.9. Essentially the arguments from Lemma 1.3.16 show that a bounded below complex of projective/flat R-modules is semiprojective/semiflat. A dual argument can be used to verify that a bounded above complex of injectives is semiinjective. Hence projective (or injective) resolutions of a module are examples of semiprojective (or semiinjective, respectively) R-complexes.

ii) *Remark* 1.3.10. As semiprojectivity of P requires the underlying chain level functor $\mathrm{Hom}_R(P,-)$ to be exact it follows that each P_i is a projective R-module for all $i\in\mathbb{Z}$. [Analogous statements hold for semiinjective and semiflat modules.] For all three, degreewise inj./flat + preserves $C\simeq 0$.

iii) An R-complex P is semiprojective if and only if it has the following lifting property: Given the solid arrows below, there is a lift making the following diagram commute

$$\begin{array}{ccc} & & X \\ & \overset{\exists}{\nearrow} & \downarrow\simeq \\ P & \longrightarrow & Y \end{array}$$

where the lift is unique up to homotopy. It is a straightforward exercise to write down and prove the corresponding lifting property for semiinjective R-complexes. It follows from this lifting property, [that a quasi-isomorphism between semiprojective (or semiinjective, respectively) R-complexes is a homotopy equivalence.] By looking at the diagram.

Finally, projective/injective objects in the category of R-complexes are necessarily contractible.

The main theorem of the section is the following. so semiproj./semiinj. are not proj./inj.

④ **Theorem 1.3.11.** *For each R-complex M there is a surjective quasi-isomorphism $P\to M$ and an injective quasi-isomorphism $M\to I$ where P is semiprojective and I is semiinjective. The R-complexes P and I are unique up to homotopy equivalence, respectively.*

The map $P\to M$ (or simply P itself) in Theorem 1.3.11 is called a **semiprojective resolution** of M and the map $M\to I$ (or simply I itself) is called a **semiinjective resolution** of M. To prove both of these results, we need a suitable notion of freeness for complexes. The first part of the theorem, regarding existence of semiprojective resolutions, is obtained by combining Lemmas 1.3.13 and 1.3.16. The uniqueness follows from the lifting properties discussed in Remark 1.3.10. The existence of semiinjective resolutions takes a bit more work and is given at the end of section; cf. Lemmas 1.3.19 and 1.3.20. semifree/semicofree.

Definition 1.3.12. An R-complex F is **semifree** if it admits a filtration by subcomplexes

$$0=F(-1)\subseteq F(0)\subseteq F(1)\subseteq F(2)\subseteq\ldots\subseteq F \qquad \text{equivalently, } F(i) \text{ is free.}$$

where $\operatorname{colim} F(i)=F$ and each subquotient $F(i)/F(i-1)$ is a sum of shifts of R (that is to say, a free graded R-module with trivial differential).

i) **Lemma 1.3.13.** *Each R-complex has a semifree resolution.* the first step relies on $Z(M)$.

Proof. Fix an R-complex M, and pick a generating set Z^0 for $\mathrm{Z}(M)$. Define $F(0)=\bigoplus_{z\in Z^0}\Sigma^{|z|}R$ and $\varepsilon(0)\colon F(0)\to M$ by $\Sigma^{|z|}1\mapsto z$. Note that $\mathrm{H}(\varepsilon(0))\colon F(0)\to \mathrm{H}(M)$ is surjective as the map on cycles is surjective.

Assuming we have constructed $F(0)\subseteq F(1)\subseteq\ldots\subseteq F(i)$ with $F(j)/F(j-1)$ a direct sum of shifts of R and compatible maps $\varepsilon(j)\colon F(j)\to M$ extending $\varepsilon(0)$, we outline the construction of the next step of the

filtration $F(i+1)$. Fix a set of cycles Z^i in $F(i)$ whose homology classes form a generating set for $\ker \mathrm{H}(\varepsilon(i))$, and note that for each $z \in Z^i$ there exists $m_z \in M$ such that

$$\partial^M(m_z) = \varepsilon(i)(z)\,. \tag{1.3.13.1}$$

Now define

$$F(i+1) := \operatorname{cone}\left(\bigoplus_{z\in Z^i} \Sigma^{|z|}R \xrightarrow{\oplus \Sigma^{|z|}1\mapsto z} F(i-1)\right)$$

attaching $|Z^i|$ copies of R

and $\varepsilon(i+1)\colon F(i+1) \to M$ given by

$$\left(f, \sum_{\text{finite}} (\Sigma^{|z|+1}r)\right) \mapsto \varepsilon(i)(f) + \sum_{\text{finite}} rm_z\,;$$

it follows from (1.3.13.1) that $\varepsilon(i+1)$ is map of R-complexes extending $\varepsilon(i)$. Furthermore, as $F(i+1)$ is a mapping cone, the quotient $F(i+1)/F(i)$ is a direct sum of shifts of R (namely, it is $\bigoplus_{z\in Z^i} \Sigma^{|z|+1}R$). This completes the inductive step of the construction.

Now set $F := \operatorname{colim} F(i)$ and $\varepsilon := \operatorname{colim} \varepsilon(i)\colon F \to M$. By construction, F is semifree, $\mathrm{H}(\varepsilon)$ is surjective (since it extends $\varepsilon(0)$), and $\ker \mathrm{H}(\varepsilon)$ is zero by the choice of cycles in the inductive step. Therefore, ε is a quasi-isomorphism and so it remains to use the exact sequences involving a complex, its boundaries, its cycles and its homology and the already justified facts that ε is surjective on cycles and homology to get that it is a surjective map of complexes. □

Remark 1.3.14. By analyzing the proof of Lemma 1.3.13, it follows that when R is noetherian and $\mathrm{H}(M)$ is finitely generated over R, then there exists a semifree resolution $F \to M$ such that each F_i is a finite rank free R-module and F is bounded below (in fact, one can take $F = F_{\geqslant i}$ for $i = \inf\{j \in \mathbb{Z} : \mathrm{H}_j(M) \neq 0\}$).

ii)

Remark 1.3.15. Given an inverse system of R-complexes

$$\dots X^2 \xrightarrow{\delta^2} X^1 \xrightarrow{\delta^1} X^0 \to 0\,,$$

there is an exact sequence

$$0 \to \varprojlim X^i \to \prod_{i\geqslant 0} X^i \xrightarrow{d^X} \prod_{i\geqslant 0} X^i$$

$\lim^1 := \operatorname{coker} d^X$ is 0

with d^X the map $(x^i)_{i\in\mathbb{Z}} \mapsto (x^i - \delta^{i+1}(x^{i+1}))_{i\in\mathbb{Z}}$. Furthermore, when the system is surjective (i.e., each δ^i is surjective), then d^X is surjective. In particular, it follows that the inverse limit of a surjective system of acyclic complexes is again acyclic.

since $\prod$ is exact, $\prod X^i \simeq$ if $X^i \simeq 0\ \forall i$.

iii)

Lemma 1.3.16. *A semifree R-complex is semiprojective. A semiprojective R-complex is semiflat.*

Proof. For the first statement, fix a semifree R-complex F with semifree filtration

$$0 = F(-1) \subseteq F(0) \subseteq F(1) \subseteq F(2) \subseteq \dots \subseteq F$$

Let A be an acyclic complex. As F is a complex of free R-modules, it suffices to show $\operatorname{Hom}_R(F, A)$ is acyclic. By induction, and using that $0 \to F(i-1) \to F(i) \to F(i)/F(i-1) \to 0$ is split exact (when forgetting the differentials), it follows that $\operatorname{Hom}_R(F(i), A) \simeq 0$.

free, so $\operatorname{Hom}(F,A) = A^{\wedge} \simeq 0$

Again using the splitting of $F(i-1) \to F(i)$ as graded R-modules, it follows that the following system is surjective (*)

$$\dots \to \operatorname{Hom}_R(F(2), A) \to \operatorname{Hom}_R(F(1), A) \to \operatorname{Hom}_R(F(0), A)\,;$$

also, since each term in the inverse system is acyclic, then by Remark 1.3.15, we obtain the first quasi-isomorphism below

$$0 \simeq \varprojlim \operatorname{Hom}_R(F(i), A) \cong \operatorname{Hom}_R(\operatorname{colim} F(i), A) = \operatorname{Hom}_R(F, A)\,,$$

(*) Using the splitting means $\exists\, r\colon F(i) \to F(i-1)$, and for any $\varphi \in \operatorname{Hom}_R(F(i-1), A)$, $u^*(\varphi r) = \varphi r u = \varphi$, so u^* is surjective.

as needed.

The second statement is left as an exercise.[3] □

Notation 1.3.17. For the remainder of the section, let E denote the injective cogenerator $\operatorname{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$ for R, and set $(-)^\vee := \operatorname{Hom}_R(-, E)$.

Definition 1.3.18. An R-complex J is **semicofree** if it admits a filtration by subcomplexes

$$J = J(0) \supseteq J(1) \supseteq J(2) \supseteq \dots$$

where $\varprojlim J/J(i) \cong J$ and each subquotient $J(i)/J(i+1) \cong (F^i)^\vee$ with F^i a semifree R-complex.

Lemma 1.3.19. *A semicofree R-complex is semiinjective.*

Proof. Let A be an acyclic complex, and note that since J is a complex of injective R-modules it suffices to show $\operatorname{Hom}_R(A, J)$ is acyclic. To see this, observe that

$$\operatorname{Hom}_R(A, J) = \operatorname{Hom}_R(A, \varprojlim J/J(i)) \cong \varprojlim \operatorname{Hom}_R(A, J/J(i)),$$

and so it suffices to show each $\operatorname{Hom}_R(A, J/J(i))$ is acyclic; cf. Remark 1.3.15. This is again an induction argument using the split exact of graded R-module $0 \to J(i-1)/J(i) \to J/J(i) \to J/J(i-1) \to 0$. □

Lemma 1.3.20. *Every R-complex admits a semicofree resolution.*

Proof. By Lemma 1.3.13, there exists a semifree resolution $F^0 \xrightarrow{\simeq} M^\vee$, and dualizing yields an injective map

$$M \to M^{\vee\vee} \xrightarrow{\simeq} (F^0)^\vee.$$

Set $K^0 = (F^0)^\vee$, and consider the exact sequence of R-complexes

$$0 \to M \to K^0 \to M^0 \to 0,$$

then again apply Lemma 1.3.13 to obtain a semifree resolution $F^1 \xrightarrow{\simeq} (M^0)^\vee$. Dualizing this map yields a left-exact sequence of R-complexes

$$0 \to M \to K^0 \to K^1$$

with $K^1 = (F^1)^\vee \xleftarrow{\simeq} (M^0)^{\vee\vee}$. Continuing in this fashion we obtain a complex of R-complexes

$$0 \to M \xrightarrow{\iota} K^0 \xrightarrow{\delta^0} K^1 \xrightarrow{\delta^1} K^2 \to \dots$$

where using properties of $(-)^\vee$ it follows that the induced sequence of graded R-modules in homology is exact:

$$0 \to \mathrm{H}(M) \to \mathrm{H}(K^0) \to \mathrm{H}(K^1) \to \mathrm{H}(K^2) \to \dots. \tag{1.3.20.1}$$

Now set $J = \operatorname{Tot}^\Pi(K^0 \to K^1 \to K^2 \to K^3 \to \dots)$, and let

$$J(i) := \operatorname{Tot}^\Pi(\dots \to 0 \to K^i \to K^{i+1} \to K^{i+2} \to \dots).$$

Then $J(i)/J(i+1) \cong K^i \cong (F^i)^\vee$ with F^i semifree, and $J \cong \varprojlim J/J(i)$. Furthermore, as ι is injective it follows that $\alpha\colon M \to J$ is injective, and so it remains to check α is a quasi-isomorphism.

To this end, observe that

$$C := \operatorname{cone}\alpha = \operatorname{Tot}^\Pi\left(0 \to M \xrightarrow{\iota} K^0 \xrightarrow{\delta^0} K^1 \xrightarrow{\delta^1} K^2 \to \dots\right)$$

[3] This is Exercise 2 on Discussion Problems 1.

and let $z = (\Sigma m, x^0, \Sigma^{-1}x^1, \Sigma^{-2}x^2, \dots)$ be a cycle in C. In particular,

$$\partial^M(m) = 0\,, \quad \iota(m) + \partial^{X^0}(x^0) = 0 \quad \text{and} \quad \delta^n(x^n) + (-1)^{n+1}\partial^{X^n}(x^n) = 0 \text{ for } n \geqslant 0. \tag{1.3.20.2}$$

Now using that $\mathrm{H}(\iota)([m]) = [\iota(m)] = 0$ in $\mathrm{H}(X^0)$ it follows from the exactness of (1.3.20.1) that $[m] = 0$ in $\mathrm{H}(M)$ and so there exists n such that $\partial^M(n) = m$. Now consider the element $x^0 + \iota(n) \in X^0$. Observe that

$$\partial^{X^0}(x^0 + \iota(n)) = \partial^{X^0}(x^0) + \partial^{X_0}(\iota(n)) = -\iota(m) + \iota(\partial^M(n)) = 0\,,$$

and hence, $x^0 + \iota(n)$ is a cycle in X^0. Also,

$$\delta^0(x^0 + \iota(n)) = \delta^0(x^0) = \partial^{X^1}(x^1)$$

and hence, $[x^0 + \iota(n)] \in \ker \mathrm{H}(\delta^0)$ and so from exactness of (1.3.20.1) it follows that $[x^0 + \iota(n)] = [\iota(n')]$ for some $n' \in \mathrm{Z}(M)$. That is to say,

$$x^0 = \iota(n' - n) + \partial^{X^0}(y^0) \quad \text{and} \quad \partial^{\Sigma M}(n' - n) = -\partial^M(-n) = m\,.$$

Next look at $x^1 - \delta^0(y^0)$ in X^1 and consider

$$\partial^{X^1}(x^1 - \delta^0(y^0)) = \delta^{X^0}(x^0) - \delta^{X^0}(x^0) + \delta^{X^0}(\iota(n' - n)) = 0$$

and so $x^1 - \delta^0(y^0)$ is a cycle. Moreover,

$$\delta^1(x^1 - \delta^0(y^0)) = \delta^{X^1}(x^1) = \partial^{X^2}(x^2)$$

by (1.3.20.2), and hence using (1.3.20.1) it follows that there is a cycle $(y^0)'$ in X^0 such that

$$x^1 = \delta^0((y^0)' + y^0) - \partial^{X^1}(y^1) \quad \text{and} \quad x^0 = \iota(n' - n) + \partial^{X^0}((y^0)' + y^0)\,.$$

Continuing in this fashion, we can find

$$y = (\Sigma(n' - n), (y^0)' + y^0, \Sigma^{-1}((y^1)' - y^1), \dots)$$

with $\partial(y) = z$, as needed. □

(5)

Definition 1.3.21. For each R-complex M, define $\mathrm{RHom}_R(M, -)$ and $M \otimes^{\mathrm{L}}_R -$ by $\mathrm{Hom}_R(P, -)$ and $P \otimes_R -$, respectively, where $P \xrightarrow{\simeq} M$ is a semiprojective resolution of M.

since P semiproj. & semiflat, preserves $\simeq$.

Remark 1.3.22. By definition of semiprojective, as well as Lemma 1.3.16, it follows that $\mathrm{RHom}_R(M, -)$ and $M \otimes^{\mathrm{L}}_R -$ are well-defined functors on $\mathsf{D}(R)$. Furthermore, as any choice of P is unique up to homotopy equivalence, we obtain isomorphism models for these functors. Furthermore, if $F \xrightarrow{\simeq} N$ with F semiflat and $N \xrightarrow{\simeq} I$ with I semiinjective, it follows

$$M \otimes^{\mathrm{L}}_R N = P \otimes_R N \xleftarrow{\simeq} P \otimes_R F \xrightarrow{\simeq} M \otimes_R F$$

and

$$\mathrm{RHom}_R(M, N) = \mathrm{Hom}_R(P, N) \xrightarrow{\simeq} \mathrm{Hom}_R(P, I) \xleftarrow{\simeq} \mathrm{Hom}_R(M, I)\,.$$

Therefore, $-\otimes^{\mathrm{L}}_R N$ and $\mathrm{RHom}_R(-, N)$ can also be computed by replacing N with a quasi-isomorphic semiflat R-complex for the first functor and replacing N with a quasi-isomorphic semiinjective R-complex for the second functor. to compute $\mathrm{RHom}_R(M,N)$, we can also replace N with semiinj. I.

Definition 1.3.23. For R-complexes M and N, define

$$\mathrm{Tor}^R(M, N) := \mathrm{H}(M \otimes^{\mathrm{L}}_R N) \quad \text{and} \quad \mathrm{Ext}_R(M, N) := \mathrm{H}(\mathrm{RHom}_R(M, N))\,.$$

(6)

1.3.24. Recall a full subcategory S of a triangulated category T is **thick** it is a triangulated subcategory that is closed under direct summands. That is, the following hold:

1. if X is in S, then so is ΣX;
2. if $X \to Y \to Z \to$ is exact and X, Y belong to S, then Z is in S;
3. if $X = X' \oplus X''$ and X is in S, then X', X'' are also in S.

} triangulated subcat.

If X is an object of T, then we let $\mathsf{thick}(X)$ denote the smallest thick sucbategory of T containing X; as intersections of thick subcategories are again thick, this category (called the *thick closure of* X) is well-defined.

Example 1.3.25. Let R be noetherian. Consider $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ the full subcategory of R-complexes M with $\mathrm{H}(M)$ a finitely generated R-module; this equivalent to the bounded derived category of finitely generated R-modules. As R is noetherian, this is a thick subcategory of $\mathsf{D}(R)$.

Example 1.3.26. If M is an R-complex, and consider T the full subcategory of $\mathsf{D}(R)$ consisting of all N such that $\operatorname{Tor}^R_{\gg 0}(M, N) = 0$. It is easy to check that T is a thick subcategory of $\mathsf{D}(R)$. The same is true if one replaces the Tor vanishing condition with $\operatorname{Ext}^{\gg 0}_R(M, N) = 0$ or $\operatorname{Ext}^{\gg 0}_R(N, M) = 0$, or more generally, (eventual) vanishing of some homological functor.

Definition 1.3.27. Let $\mathsf{Perf}\, R$ denote the full subcategory of $\mathsf{D}(R)$ consisting of those objects isomorphic to a bounded complex of finitely generated projective R-modules.

⊕ & direct summands preserve proj.

It is straightforward to see $\mathsf{Perf}\, R$ is a thick subcategory of $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ (and hence, also of $\mathsf{D}(R)$).

Proposition 1.3.28. *For a commutative noetherian ring R, one has* $\mathsf{Perf}\, R = \mathsf{thick}(R)$.

Proof. As $\mathsf{Perf}\, R$ is thick and R belongs to it, it follows that $\mathsf{thick}(R) \subseteq \mathsf{Perf}\, R$. For the reverse containment, consider an R-complex P having the form

$$P = 0 \to P_n \to P_{n-1} \to \ldots \to P_0 \to 0$$

where each P_i is a finitely generated projective R-module. There is an exact triangle

$$P' \to P \to \Sigma^n P_n \to$$

where $P' = 0 \to P_{n-1} \to \ldots \to P_0 \to 0$. By induction P' and $\Sigma^n P_n$ belong to $\mathsf{thick}(R)$, and so it follows that P also ~~belongs~~ belongs to $\mathsf{thick}(R)$. □

↓ since free mod. & summands in thick(R).

1.4 Local cohomology

Throughout this section, R is a commutative noetherian ring and we fix an ideal $I \subseteq R$. For an R-module M, its I-**power torsion submodule** is

$$\Gamma_I(M) := \{m \in M : I^n m = 0 \text{ for some } n \gg 0\}\,.$$

That is,

$$\Gamma_I(M) = \bigcup_{n \geqslant 0} (0 :_M I^n) = \operatorname{colim}_n \operatorname{Hom}_R(R/I^n, M)\,.$$

Recall that Γ_I is a left exact functor on $\mathsf{Mod}(R)$, and so this defines an exact functor $\mathrm{R}\Gamma_I \colon \mathsf{D}(R) \to \mathsf{D}(R)$ given by

$$M \mapsto \mathrm{R}\Gamma_I(M) := \Gamma_I(J)$$

where $M \to J$ is a semiinjective resolution of M; when M is an honest R-module, in light of Example 1.3.9, one can take J to be an injective resolution of M. As semiinjective resolutions are unique up to homotopy equivalence, it follows that $\mathrm{R}\Gamma_I(M)$ is well-defined in $\mathsf{D}(R)$, and in fact in the homotopy category of R-complexes.

Definition 1.4.1. For an R-complex M, its **local cohomology supported on** I (or $\mathrm{V}(I)$) is

$$\mathrm{H}_I(M) := \mathrm{H}(\mathrm{R}\Gamma_I(M))\,.$$

Remark 1.4.2. If $\sqrt{I} = \sqrt{I'}$, then $\mathrm{H}_I(M) \cong \mathrm{H}_{I'}(M)$ for any R-complex M as the underlying functors Γ_I and $\Gamma_{I'}$ agree from the cofinality of $\{I^n\}$ and $\{(I')^n\}$.

Remark 1.4.3. For each R-complex M, there is a map $\mathrm{R}\Gamma_I(M) \to M$ in $\mathsf{D}(R)$. Furthermore, given a map $M \to N$ in $\mathsf{D}(R)$, we get an induced map $\mathrm{R}\Gamma_I(M) \to \mathrm{R}\Gamma_I(N)$. Looking head a bit, confer Theorem 1.4.13, it follows that if M is an R-complex with I-power torsion homology, then the map $M \to N$ factors as

$$M \to \mathrm{R}\Gamma_I(N) \to N\,.$$

Definition 1.4.4. Consider a sequence of maps in $\mathsf{D}(R)$:

$$X^0 \xrightarrow{\alpha^0} X^1 \xrightarrow{\alpha^1} X^2 \xrightarrow{\alpha^2} \cdots\,.$$

A **homotopy colimit** of the sequence, denoted $\operatorname{hocolim}(X^i)$, is defined by the triangle

$$\bigoplus_{i\geqslant 0} X^i \xrightarrow{1-\alpha} \bigoplus_{i\geqslant 0} X^i \to \operatorname{hocolim}(X^i) \to\,;$$

here the map $1-\alpha$ is determined by the maps on the components given by

$$X^i \xrightarrow{\begin{bmatrix}\mathrm{id}\\ -\alpha^i\end{bmatrix}} X^i \oplus X^{i+1} \hookrightarrow \bigoplus_{i\geqslant 0} X^i\,.$$

by triangulated : cone is uniq.

Remark 1.4.5. Given a sequence of maps $X^0 \xrightarrow{\alpha^0} X^1 \xrightarrow{\alpha^1} X^2 \xrightarrow{\alpha^2} \cdots$ in $\mathsf{D}(R)$, then [a homotopy colimit of the sequence is unique up to a non-unique isomorphism in $\mathsf{D}(R)$.] Furthermore, applying homology defines directed system of graded R-modules

$$\mathrm{H}(X^0) \xrightarrow{\mathrm{H}(\alpha^0)} \mathrm{H}(X^1) \xrightarrow{\mathrm{H}(\alpha^1)} \mathrm{H}(X^2) \xrightarrow{\mathrm{H}(\alpha^2)} \cdots\,,$$

which is independent of choice of representatives for the sequence in $\mathsf{D}(R)$, and we have the following isomorphism of graded R-modules

$$\mathrm{H}(\operatorname{hocolim} X^i) \cong \operatorname{colim}_i \mathrm{H}(X^i)\,;$$

in particular, the homology of homotopy colimits is [unique up to canonical isomorphisms of graded R-modules.] since it's real colim.

1.4.6. Fix an R-complex M. The natural maps $R/I^{n+1} \to R/I^n$ define commutative diagrams in $\mathsf{D}(R)$:

$$\begin{array}{ccc} \mathrm{RHom}_R(R/I^n, M) & \longrightarrow & \mathrm{RHom}_R(R/I^{n+1}, M) \\ & \searrow & \downarrow \\ & & \mathrm{R}\Gamma_I(M)\,. \end{array}$$

The following is now a rephrasing of the definition of local cohomology.

$\operatorname{colim}\mathrm{Hom}_R(R/I^n, J)$
$=$

Proposition 1.4.7. *For any R-complex M, [there is an isomorphism* $\operatorname{hocolim}\mathrm{RHom}_R(R/I^n, M) \to \mathrm{R}\Gamma_I(M)$ *in* $\mathsf{D}(R)$.*] In particular, there is an isomorphsim of graded R-modules* $\operatorname{colim}_n \operatorname{Ext}_R(R/I^n, M) \cong \mathrm{H}_I(M)$.

Notation 1.4.8. For a sequence of element $\boldsymbol{x}$ in R, we write $K(\boldsymbol{x})$ for the Koszul complex on $\boldsymbol{x}$ over R. For an R-complex M, we write $K(\boldsymbol{x}; M) = K(\boldsymbol{x}) \otimes_R M$ and $\mathrm{H}(\boldsymbol{x}; M) := \mathrm{H}(K(\boldsymbol{x}; M))$.

1.4.9. Now fix a generating set $\boldsymbol{x} = x_1, \dots, x_c$ for I. Consider $\boldsymbol{x}^n := x_1^n, \dots, x_c^n$ for each $n \geqslant 1$, and set $I_n := (\boldsymbol{x}^n)$. The maps of R-complexes $K(\boldsymbol{x}^n) \to R/I_n$ define maps

$$\mathrm{RHom}_R(R/I_n, M) \to \mathrm{RHom}_R(K(\boldsymbol{x}^n), M)$$

in $\mathsf{D}(R)$, for each R-complex M. This defines a map

$$\mathrm{R}\Gamma_I(M) \simeq \operatorname{hocolim} \mathrm{RHom}_R(R/I_n, M) \to \operatorname{hocolim} \mathrm{RHom}_R(K(\boldsymbol{x}^n), M) \simeq \operatorname{colim}_n \mathrm{Hom}_R(K(\boldsymbol{x}^n), R) \otimes_R M\,.$$

We want to analyze the right-hand side.

First, note that there is an isomorphism of R-complexes

$$\mathrm{Hom}_R(K(\boldsymbol{x}^n), R) \cong \Sigma^{-c} K(\boldsymbol{x}^n) = \bigotimes_{i=1}^{c} \Sigma^{-1} K(x_i^n) \cong \bigotimes_{i=1}^{c} \mathrm{Hom}_R(K(x_i^n), R)$$

and there is the corresponding commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}_R(K(\boldsymbol{x}^n), R) & \xrightarrow{\cong} & \bigotimes_{i=1}^{c} \mathrm{Hom}_R(K(x_i^n), R) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(K(\boldsymbol{x}^{n+1}), R) & \xrightarrow{\cong} & \bigotimes_{i=1}^{c} \mathrm{Hom}_R(K(x_i^{n+1}), R) \end{array}$$

where the map on the right is the tensor product of maps dual to $K(x_i^{n+1}) \to K(x_i^n)$. In particular, we have

$$\operatorname{colim}_n \mathrm{Hom}_R(K(\boldsymbol{x}^n), R) \cong \bigotimes_{i=1}^{c} \operatorname{colim}_n \mathrm{Hom}_R(K(x_i^n), R)\,;$$

as a matter of notation let $K_\infty(\boldsymbol{x}) = \operatorname{colim}_n \mathrm{Hom}_R(K(\boldsymbol{x}^n), R)$, called the **stable Koszul complex on $\boldsymbol{x}$**, then the isomorphism above asserts that as R-complexes we have the isomorphism $K_\infty(\boldsymbol{x}) \cong \bigotimes_{i=1}^{c} K_\infty(x_i)$. Hence, to determine $K_\infty(\boldsymbol{x})$ we can consider the case when $c = 1$.

Observe that $K_\infty(x)$ is the colimit of the diagram

$$\begin{array}{lccccccc} \text{degree } 0: & R & = & R & = & R & = & \cdots \\ & \downarrow{\scriptstyle x} & & \downarrow{\scriptstyle x^2} & & \downarrow{\scriptstyle x^3} & & \\ \text{degree } -1: & R & \xrightarrow{x} & R & \xrightarrow{x} & R & \xrightarrow{x} & \cdots \end{array}$$

where the columns are the dual Koszul complexes $\mathrm{Hom}_R(K(x^n), R)$, for $n = 1, 2 \dots$, and the horizontal maps are the maps of complexes induced by obvious ones $K(x^{n+1}) \to K(x^n)$. It then follows that

$$K_\infty(x) \cong 0 \longrightarrow R \longrightarrow R_x \longrightarrow 0$$

where R is in degree 0 and the differential is the localization map.

For a general c, the isomorphism $K_\infty(\boldsymbol{x}) = \operatorname{colim}_n \mathrm{Hom}_R(K(\boldsymbol{x}^n), R)$ combined with the analysis above yields

$$K_\infty(\boldsymbol{x}) = 0 \to R \to \bigoplus_{i=1}^{c} R_{x_i} \to \bigoplus_{i<j} R_{x_i x_j} \to \ldots \to R_{x_1 \cdots x_c} \to 0$$

where R is in degree 0, where the maps are (signed) sums of localization maps (determined from the tensor product differential). Furthermore, in summary, for any R-complex M, there is a map in $\mathsf{D}(R)$:

$$\mathrm{R}\Gamma_I(M) \to K_\infty(\boldsymbol{x}) \otimes_R M$$

where $\boldsymbol{x}$ is a generating set for I.

Theorem 1.4.10. *For any R-complex M, the natural map $\mathrm{R}\Gamma_I(M) \to K_\infty(\boldsymbol{x}) \otimes_R M$ is an isomorphism in $\mathsf{D}(R)$, where $\boldsymbol{x}$ is a generating set for I.*

Proof. One can replace M with a semiinjective resolution to verify the quasi-isomorphism. A spectral sequence argument reduces the problem to showing that $\mathrm{R}\Gamma_I(E) \xrightarrow{\simeq} K_\infty(\boldsymbol{x}) \otimes_R E$ for each injective R-module E, and using the structure theorem on injectives we can assume $E = E_R(R/\mathfrak{p})$ for some prime $\mathfrak{p}$ of R. Next, recall that

$$\mathrm{R}\Gamma_I(E) \simeq \Gamma_I(E) = \begin{cases} E & \mathfrak{p} \supseteq I \\ 0 & \text{otherwise}\,, \end{cases}$$

and so it suffices to establish $K_\infty(\boldsymbol{x}) \otimes_R E \simeq \Gamma_I(E)$.

Now note that induction on c, gives that for each $i, n > 0$ there exists $j > 0$ such that

$$\mathrm{H}_i(K(\boldsymbol{x}^{n+j})) \to \mathrm{H}_i(K(\boldsymbol{x}^n)) \text{ is zero;} \tag{1.4.10.1}$$

for the base case, the maps above can be identified with

$$(0 : x^{n+j}) \xrightarrow{x^j \cdot} (0 : x^n)$$

while the noetherianity of R gives that $(0 : x^{n+j}) = (0 : x^{n+j+1}) = \dots$ for $j \gg 0$, and so the maps above vanish for $j \gg 0$, as claimed.

Hence, whenever $i > 0$ we have

$$\begin{aligned} \mathrm{H}_{-i}(K_\infty(\boldsymbol{x}) \otimes_R E) &\cong \mathrm{H}_{-i}(\operatorname{colim}_n \operatorname{Hom}_R(K(\boldsymbol{x}^n), E)) \\ &\cong \operatorname{colim}_n \operatorname{Hom}_R(\mathrm{H}_i(K(\boldsymbol{x}^n)), E)) \\ &= 0 \end{aligned}$$

where the first isomorphism used an isomorphism from 1.4.9, the second uses that filtered colimits commute with homology and that E is injective, and the third uses (1.4.10.1). Therefore, $K_\infty(\boldsymbol{x}) \otimes_R E \simeq \mathrm{H}_0(K_\infty \otimes_R E)$, and now finally it remains to notice that

$$\mathrm{H}_0(K_\infty \otimes_R E) = \ker(E \to \bigoplus_{i=1}^{c} E_{x_i}) = \Gamma_I(E)\,,$$

with the second equality following from the fact that $\boldsymbol{x}$ generates I, finishing the proof of the theorem. □

Now we record some basic facts on local cohomology; some of these being immediate corollaries of Theorem 1.4.10. Furthermore, let $\operatorname{ara}(I)$ denote the **arithmetic rank** of I; this is the least nonnegative integer that I is generated by up to radical.

Proposition 1.4.11. *Let M be an R-complex.*

(a) $\mathrm{H}_I(M)$ *is I-power torsion.*
(b) $\mathrm{H}_I(-)$ *commutes with arbitrary direct sums.*
(c) *If M is an R-module, then $\mathrm{H}^i_I(M) = 0$ for $i > \operatorname{ara}(I)$.*
(d) *If $R \to S$ is a ring map and N is an S-complex, then $\mathrm{H}_I(N) \cong \mathrm{H}_{IS}(N)$.*
(e) *If $R \to S$ is a flat ring map, then $\mathrm{H}_I(M) \otimes_R S \cong \mathrm{H}_I(M \otimes_R S) \cong \mathrm{H}_{IS}(M \otimes_R S)$.*

We isolate two specializations of the facts above.

Corollary 1.4.12. (a) *If $(R, \mathfrak{m})$ is local and M is an R-module, then $\mathrm{H}^i_{\mathfrak{m}}(M) = 0$ for $i > \dim(R)$.*

(b) Let $S = U^{-1}R$ in 1.4.11(e).

(b) *If U is a multiplicatively closed subset of R, then $U^{-1}\operatorname{H}_I(M) \cong \operatorname{H}_I(U^{-1}M) \cong \operatorname{H}_{U^{-1}I}(U^{-1}M)$ for any R-complex M.*

Theorem 1.4.13. *The natural map $\mathrm{R}\Gamma_I(M) \to M$ is an isomorphism in $\mathsf{D}(R)$ if and only if $\operatorname{H}(M)$ is I-power torsion.*

H(M) ≅ $H_I(M)$ → I-power torsion.

Proof. The forward direction is Proposition 1.4.11(a). For the converse, again use the Künneth spectral sequence

$$\mathrm{E}^2_{p,q} = \operatorname{H}^p_I(\operatorname{H}_{-q}(M)) \implies \operatorname{H}^{p+q}_I(M)$$

and so $\mathrm{E}^2_{p,q} = 0$ whenever $p \neq 0$; to see this note that

$$K_\infty(\boldsymbol{x})_p \otimes_R \operatorname{H}_{-q}(M) = 0 \quad \text{when } p \neq 0\,,$$

with $\boldsymbol{x}$ generating I, since $\operatorname{H}_{-q}(M)_{x_i} = 0$ for each i where $p < 0$, and $K_\infty(\boldsymbol{x})_p = 0$ when $p > 0$. Finally it remains to observe that

$$\mathrm{E}^2_{0,q} = \Gamma_I(\operatorname{H}_{-q}(M)) = \operatorname{H}_{-q}(M)$$

and so we obtain isomorphisms $\operatorname{H}_{-q}(M) \xrightarrow{\cong} \operatorname{H}^q_I(M)$ for each q, and so $\mathrm{R}\Gamma_I(M) \to M$ is an isomorphism. □

1.4.14. Fix R-complexes M, N and consider the natural map

$$\operatorname{RHom}_R(M, \mathrm{R}\Gamma_I(N)) \to \operatorname{RHom}_R(M, N)$$

induced by $\mathrm{R}\Gamma_I(N) \to N$. As the left-hand complex has I-power torsion homology, it follows that the map above induces a map

(in D(R)) $\qquad \operatorname{RHom}_R(M, \mathrm{R}\Gamma_I(N)) \to \mathrm{R}\Gamma_I(\operatorname{RHom}_R(M, N))\,.$ apply 1.4.13.

Proposition 1.4.15. *The natural map $\operatorname{RHom}_R(M, \mathrm{R}\Gamma_I(N)) \to \mathrm{R}\Gamma_I(\operatorname{RHom}_R(M, N))$ is an isomorphism in $\mathsf{D}(R)$ in the following cases:*

(a) *For any N when M is in $\mathsf{Perf}\, R$;*

(b) *when M is in $\mathsf{D}^{\mathsf{b}}(\mathit{mod}\, R)$ and $\operatorname{H}_i(N) = 0$ for $i \gg 0$.*

Proof. We only prove (a) as (b) is an exercise.[4] For (a), if one consider the category of all L in $\mathsf{D}(R)$ where

$$\operatorname{RHom}_R(L, \mathrm{R}\Gamma_I(N)) \xrightarrow{\simeq} \mathrm{R}\Gamma_I(\operatorname{RHom}_R(L, N))$$

is an isomorphism, then this is a thick subcategory. It is easy to see that R belongs to this category and hence, so must $\mathsf{Perf}\, R$; this finishes the proof of (a). □

thick(R) ⊂ this category.

1.5 Depth

In this section R is a commutative noetherian ring and I is an ideal of R.

Definition 1.5.1. For an R-complex M, define

$$\inf M := \inf\{i \in \mathbb{Z} : \operatorname{H}_i(M) \neq 0\} \quad \text{and} \quad \sup M := \sup\{i \in \mathbb{Z} : \operatorname{H}_i(M) \neq 0\}\,.$$

Definition 1.5.2. The I-**depth** of an R-complex M is

$$\operatorname{depth}_I(M) = -\sup \operatorname{RHom}_R(R/I, M) = \inf\{i \in \mathbb{Z} : \operatorname{Ext}^i_R(R/I, M) \neq 0\}\,.$$

[4] This is Exercise 3 on Discussion Problems 1.

$\operatorname{Ext}^i_R = H^i(\mathrm{RHom}) = -H_i(\mathrm{RHom})$.

$K(\underline{x}; M) \simeq N$

Theorem 1.5.3. *Suppose an ideal I of R is generated by $\boldsymbol{x} = x_1, \dots, x_n$. For any R-complex M, one has*

$$\operatorname{depth}_I(M) = n - \sup K(\boldsymbol{x}; M) = -\sup \mathrm{R}\Gamma_I(M)\,.$$

That is, $\operatorname{depth}_I(M) = n - \sup\{i \in \mathbb{Z} : \mathrm{H}_i(\boldsymbol{x}; M) \neq 0\} = \inf\{i \in \mathbb{Z} : \mathrm{H}^i_I(M) \neq 0\}$.

Remark 1.5.4. Recall a sequence elements $\boldsymbol{x}$ in R is **Koszul regular** if $K(\boldsymbol{x}) \xrightarrow{\simeq} R/(\boldsymbol{x})$. More generally, $\boldsymbol{x}$ is M**-Koszul regular** if $K(\boldsymbol{x}) \otimes_R M \xrightarrow{\simeq} M/(\boldsymbol{x})M$. Recall that for an R-module M, the sequence $\boldsymbol{x}$ is M**-regular** if $(\boldsymbol{x})M \neq M$ and $M/(x_1, \dots, x_{i-1}) \xrightarrow{x_i \cdot} M/(x_1, \dots, x_{i-1})$ is injective for each i. It is straightforward to verify that an M-regular sequence is always M-Koszul regular, and the converse holds when $\boldsymbol{x}$ is contained in the Jacobson radical of R.

Given a nonzero finitely generated R-module M, it is a classical argument that the length of a maximal M-Koszul regular sequence is independent of choice of the sequence and is exactly $\operatorname{depth}_I(M)$. In particular, when I is contained in the Jacobson radical of R, it follows that $\operatorname{depth}_I(M)$ is the length of the longest M-regular sequence.

Before starting on the proof of Theorem 1.5.3, we need some preliminary remarks and a lemma. The first fact below is easy to verify and the second follows from the first.

Remark 1.5.5. Fix R-complexes X, Y.

(a) $\sup \mathrm{RHom}_R(X, Y) \leqslant \sup Y - \inf X$.

(b) If $\mathrm{H}(X)$ is bounded, then

$$\sup \mathrm{RHom}_R(X, Y) \leqslant \sup\{\sup \mathrm{RHom}_R(\mathrm{H}_i(X), Y) - i : i \in \mathbb{Z}\}\,.$$

since for $i \geq \inf X$, $K_i \geq 0$:
since $Y \to J$ has neg. deg. $\leftarrow$ for negative degree, RHom = 0.
$\leq \sup \mathrm{RHom}_R(\mathrm{H}_i(X), Y) - \inf X$
K_i

Lemma 1.5.6. *If $\mathrm{H}(X)$ is bounded and $I^t \mathrm{H}(X) = 0$ for some $t > 0$, then for any R-complex M:*

$$\sup \mathrm{RHom}_R(X, M) \leqslant \sup \mathrm{RHom}_R(R/I, M) - \inf X\,.$$

Proof. By Remark 1.5.5(b), we can assume X is concentrated in degree zero and so X is an R-module with $I^t X = 0$. Now observe that

since suffices to show for $\mathrm{H}_i(X)$: See 1.5.5(b).
$\mathrm{H}(X) = X$

$$\mathrm{RHom}_R(X, M) \simeq \mathrm{RHom}_{R/I^t}(X, \mathrm{RHom}_R(R/I^t, M))$$

f lands in I^t-torsion.

and so applying Remark 1.5.5(a) we have

$$\sup \mathrm{RHom}_R(X, M) \leq \sup \mathrm{RHom}_R(R/I^t, M) - \inf X = \sup \mathrm{RHom}_R(R/I^t, M)\,.$$

$= 0$

So we inductively show $\sup \mathrm{RHom}_R(R/I^n, M) \leq \sup \mathrm{RHom}_R(R/I, M)$; there is nothing to show for the base case. Next, consider the exact sequence

$$0 \to I^{n-1}/I^n \to R/I^n \to R/I^{n-1} \to 0$$

which determines the exact triangle

$:= A$

$\sup \mathrm{RHom}_R(I^{n-1}/I^n, \mathrm{RHom}(R/I, M)) \leq \sup A - \cdots \leq \sup A$

$$\mathrm{RHom}_R(R/I^{n-1}, M) \to \mathrm{RHom}_R(R/I^n, M) \to \mathrm{RHom}_R(I^{n-1}/I^n, M) \to\,.$$

The desired result now follows, from induction, as the supremum of the outer two complexes is at most $\sup \mathrm{RHom}_R(R/I, M)$. □

Thus $A_1 \to A_2 \to A_3$: $\sup A_2 \leq \sup\{\sup A_1, \sup A_3\}$.

Proof of Theorem 1.5.3. Set $K := K(\boldsymbol{x})$ and using the self-duality of K it follows that

$$n - \sup K(\boldsymbol{x}; M) = -\sup \mathrm{Hom}_R(K, M)\,.$$

Therefore, to prove the theorem we want to show

$$\sup \mathrm{RHom}_R(R/I, M) = \sup \mathrm{Hom}_R(K, M) = \sup \mathrm{R}\Gamma_I(M)\,.$$

For the first equality, note that Lemma 1.5.6 gives $\geqslant$. For the other inequality, consider

$$\begin{aligned}\operatorname{RHom}_R(R/I, \operatorname{Hom}_R(K, M)) &\simeq \operatorname{RHom}_R(R/I \otimes_R K, M) \\ &= \bigoplus_{i=0}^{n} \Sigma^{-i} \operatorname{RHom}_R(R/I, M)^{\oplus \binom{n}{i}}\end{aligned}$$

and combining this with Remark 1.5.5(a), we obtain the inequality below

$$\begin{aligned}\sup \operatorname{RHom}_R(R/I, M) &= \sup \left(\bigoplus_{i=0}^{n} \Sigma^{-i} \operatorname{RHom}_R(R/I, M)^{\oplus \binom{n}{i}} \right) \\ &\leqslant \sup \operatorname{Hom}_R(K, M) - \inf R/I \\ &= \sup \operatorname{Hom}_R(K, M)\,.\end{aligned}$$

For the second equality, observe that by Proposition 1.4.15(a) gives the first isomorphism below

$$\operatorname{Hom}_R(K, \mathrm{R}\Gamma_I(M)) \simeq \mathrm{R}\Gamma_I(\operatorname{Hom}_R(K, M)) \simeq \operatorname{Hom}_R(K, M)$$

and the second follows from Theorem 1.4.13. Therefore, another application of Remark 1.5.5(a) yields

$$\sup \operatorname{Hom}_R(K, M) \leqslant \sup \mathrm{R}\Gamma_I(M) - \inf K = \sup \mathrm{R}\Gamma_I(M)\,. \tag{1}$$

By Lemma 1.5.6, and the already established equality, we have

$$\sup \operatorname{RHom}_R(R/I^n, M) \leqslant \sup \operatorname{Hom}_R(K, M)$$

for each n, and so

$$\sup \mathrm{R}\Gamma_I(M) = \sup(\operatorname{hocolim} \operatorname{RHom}_R(R/I^n, M)) \leqslant \sup \operatorname{Hom}_R(K, M)\,, \tag{2}$$

completing the proof. □

For the rest of the section assume $(R, \mathfrak{m}, k)$ is local.

Definition 1.5.7. For an R-complex M, its **depth** is $\operatorname{depth}(M) = \operatorname{depth}_{\mathfrak{m}}(M)$.

1.5.8. Assume M is in $\mathsf{D}^{\mathrm{b}}(\operatorname{mod} R)$. By Nakayama's lemma it follows that M is in $\operatorname{Perf} R$ if and only if $\sup(M \otimes^{\mathrm{L}}_R k) < \infty$. We set

$$\operatorname{projdim}_R(M) := \sup(M \otimes^{\mathrm{L}}_R k) = \sup\{i \in \mathbb{Z} : \operatorname{Tor}^R_i(M, k) \neq 0\}\,;$$

another application of Nakayama's yields that $\operatorname{projdim}_R(M)$ is the smallest integer d such that

$$M \simeq (0 \to F_d \to F_{d-1} \to \dots \to F_{i+1} \to F_i \to 0)$$

with each F_j finite rank free and $i = \inf M$; confer also Remark 1.3.14.

Theorem 1.5.9 (Auslander–Buchsbaum Formula)**.** *For a local ring R, if M is in $\operatorname{Perf} R$ is nontrivial, then for any R-complex N we have*

$$\operatorname{depth}(M \otimes^{\mathrm{L}}_R N) = \operatorname{depth}(N) - \operatorname{projdim}_R(M).$$

Corollary 1.5.10. *There is an equality* $\operatorname{depth}(M) = \operatorname{depth}(R) - \operatorname{projdim}_R(M)$.

1.5.11. For R-complexes L, M, N we have a natural evaluation map

$$\operatorname{RHom}_R(L, N) \otimes_R M \to \operatorname{RHom}_R(L, N \otimes^{\mathrm{L}}_R M)\,.$$

Fixing L, N, then the subcategory of $\mathsf{D}(R)$ consisting of all M where the evaluation map is an isomorphism in $\mathsf{D}(R)$ is a thick subcategory. Moreover, it is clear that R belongs to this category and hence, the evaluation map above is any isomorphism whenever M is in $\operatorname{Perf} R$.

By Kunneth, $-\sup(A\otimes_k B) = -\sup A - \sup B$; $H_n(A\otimes B) = \bigoplus_{i+j=n} H_i(A)\otimes_k H_j(B)$ (over k-v.s.!!)

Proof of Theorem 1.5.9. Observe that there are isomorphisms

$$\operatorname{RHom}_R(k, M \otimes^{\mathrm{L}}_R N) \simeq \operatorname{RHom}_R(k, N) \otimes^{\mathrm{L}}_R M \simeq (\operatorname{RHom}_R(k, N)) \otimes_k (k \otimes^{\mathrm{L}}_R M)$$

the first one is from 1.5.11, while the second is because $\operatorname{RHom}_R(k, N)$ can be modeled by a k-complex. Now taking supremums of both sides yields the desired equality. □

On the injective side there is analogous formula relating injective dimension and depth. It is also an easy consequence of an isomorphism in the derived category; confer Lemma 1.2.3.

Definition 1.5.12. For M in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$, we define $\operatorname{inj\,dim}_R(M)$ to be the smallest integer d such that $M \xrightarrow{\simeq} J$ is a semiinjective resolution with $J_i = 0$ for $i < -d$. Equivalently,

$$\operatorname{inj\,dim}_R(M) = -\inf \operatorname{RHom}_R(k, M) = \sup\{i \in \mathbb{Z} : \operatorname{Ext}^i_R(k, M) \neq 0\}\,.$$

Let $\mathsf{InjPerf}\, R$ denote the full subcategory of $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$ consisting of those objects isomorphic to a bounded complex of injective R-modules. That is to say, $\mathsf{InjPerf}\, R$ consists exactly of the M in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$ where $\operatorname{inj\,dim}_R(M) < \infty$.

1.5.13. For R-complexes L, M, N we have a natural evaluation map

$$L \otimes^{\mathrm{L}}_R \operatorname{RHom}_R(M, N) \to \operatorname{RHom}_R(\operatorname{RHom}_R(L, M), N)\,.$$

If, for example, L, M are in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$ and N is in $\mathsf{InjPerf}\, R$, then the evaluation map above is an isomorphism. This is derived version of the evaluation isomorphism in Lemma 1.2.3; we leave verifying this version as a straightforward exercise.

Theorem 1.5.14 (Ischebeck's Theorem)**.** *For a local ring R, M in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$ and N in $\mathsf{InjPerf}\, R$, we have*

$$\sup\{i : \operatorname{Ext}^i_R(M, N) \neq 0\} = \operatorname{inj\,dim}_R(N) - \operatorname{depth}(M)\,.$$

In particular, if N is a finitely generated R-module and $\operatorname{inj\,dim}_R(N) < \infty$, then $\operatorname{inj\,dim}_R(N) = \operatorname{depth}(R)$ and

$$\sup\{i : \operatorname{Ext}^i_R(M, N) \neq 0\} = \operatorname{depth}(R) - \operatorname{depth}(M)\,.$$

Proof. The second statement follows from the first statement, by applying the first with $M = R$ to obtain $\operatorname{inj\,dim}_R(M) = \operatorname{depth}(R)$. For the first statement, observe that

$$k \otimes^{\mathrm{L}}_R \operatorname{RHom}_R(M, N) \simeq \operatorname{RHom}_R(\operatorname{RHom}_R(k, M), N) \simeq \operatorname{Hom}_k(\operatorname{RHom}_R(k, M), \operatorname{RHom}_R(k, N))\,;$$

the first isomorphism uses 1.5.13 while the second one uses adjunction. Therefore,

$$\begin{aligned}\inf \operatorname{RHom}_R(M, N) &= \inf(k \otimes^{\mathrm{L}}_R \operatorname{RHom}_R(M, N)) \\ &= \inf(\operatorname{Hom}_k(\operatorname{RHom}_R(k, M), \operatorname{RHom}_R(k, N))) \\ &= -\sup \operatorname{RHom}_R(k, M) + \inf \operatorname{RHom}_R(k, N)\,,\end{aligned}$$

where the first equality uses that $\operatorname{RHom}_R(M, N)$ is in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$, the second uses the previously established isomorphism, and the last is just bookkeeping. Now just using the definition of depth of and injective dimension, the desired result follows. □

Remark 1.5.15. For M in $\mathsf{D}^{\mathsf{b}}(\operatorname{mod} R)$ and I is an ideal of R, it is straightforward to show I-depth is local. That is to say,

$$\operatorname{depth}_I(M) = \inf\{\operatorname{depth}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) : \mathfrak{p} \in \mathrm{V}(I)\}\,.$$

In particular, when $I = 0$ we obtain

$$\sup M = \sup \operatorname{RHom}_R(R, M) = \sup\{-\operatorname{depth}(M_{\mathfrak{p}}) : \mathfrak{p} \in \operatorname{Spec}(R)\}\,.$$

1.6 Dualizing complex and local duality

Let R be a commutative noetherian ring.

Definition 1.6.1. An R-complex ω_R is a **dualizing complex** of R if:

(a) ω_R is in $\mathsf{InjPerf}\, R$;

(b) the natural map $R \xrightarrow{\simeq} \mathrm{RHom}_R(\omega_R, \omega_R)$ is an isomorphism.

Example 1.6.2. (1) Assume R is artinian local with residue field k. Then $E_R(k)$ is a dualizing complex of R.

(2) If R is a regular ring having finite Krull dimension, then R is a dualizing complex of R.

(3) If ω_R is a dualizing complex of R and U is a multiplicatively closed subset, then $U^{-1}\omega_R$ is a dualizing complex for $U^{-1}R$.

(4) Assume $R \to S$ a finite map. If ω_R is a dualizing complex of R, then $\mathrm{RHom}_R(S, \omega_R)$ is a dualizing complex of S.[5]

Remark 1.6.3. We remark that dualizing complexes may not exist. However, combining Example 1.6.2 parts (2) and (4) we obtain that any complete local ring admits a dualizing complex. In the non-local setting, they are not necessarily unique, however, when local one can speak of *the* dualizing complex of the ring; see Remark 1.6.10.

Theorem 1.6.4 (Grothendieck duality)**.** *Assume R is a commutative noetherian ring with dualizing complex ω_R. The functor $(-)^\dagger := \mathrm{RHom}_R(-, \omega_R)\colon \mathsf{D}(R) \to \mathsf{D}(R)$ restricts to self-inverse equivalences*

$$\begin{array}{ccc} \mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R) & \xrightarrow[\equiv]{(-)^\dagger} & \mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R) \\ \subseteq\uparrow & & \uparrow\subseteq \\ \operatorname{\mathsf{Perf}} R & \xrightarrow[\equiv]{(-)^\dagger} & \mathsf{InjPerf}\, R\,. \end{array}$$

Proof. First, we verify the natural map $M \to M^{\dagger\dagger}$ is an isomorphism. To this end, note that this map factors as

$$M \to M \otimes^{\mathrm{L}}_R \mathrm{RHom}_R(\omega_R, \omega_R) \xrightarrow{\simeq} \mathrm{RHom}_R(\mathrm{RHom}_R(M, \omega_R), \omega_R) = M^{\dagger\dagger}$$

where the isomorphism uses 1.5.13 (here we are making use of the fact that ω_R is in $\mathsf{InjPerf}\, R$). However, the first map in this factorization is obtain by tensoring along $R \xrightarrow{\simeq} \mathrm{RHom}_R(\omega_R, \omega_R)$ and so it is also an isomorphism; thus, $(-)^\dagger$ is a self-inverse equivalence on $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$.

Next, consider the subcategory of $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$ consisting of M with $M^\dagger$ in $\mathsf{InjPerf}\, R$; this is clearly a thick subcategory containing R as $R^\dagger = \omega_R$, and hence $(\operatorname{\mathsf{Perf}} R)^\dagger \subseteq \mathsf{InjPerf}\, R$.

Finally, if J is $\mathsf{InjPerf}\, R$ then observe that for any M in $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$ we have

$$M \otimes^{\mathrm{L}}_R J^\dagger \simeq \mathrm{RHom}_R(\mathrm{RHom}_R(M, J), \omega_R)\,;$$

again, this makes use of the evaluation isomorphism from 1.5.13. As J is in $\mathsf{InjPerf}\, R$, it follows that $\mathrm{RHom}_R(M, J)$ is $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$ and hence, $(\mathrm{RHom}_R(M, J))^\dagger$ is also in $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$. Combining this with the isomorphism above yields that

$$\mathrm{Tor}^R_{\gg 0}(M, J^\dagger) = 0 \quad \text{for all } M \text{ in } \mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)\,.$$

Therefore, since $J^\dagger$ is $\mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)$, it follows that $J^\dagger$ is in $\operatorname{\mathsf{Perf}} R$ as needed. □

[5] This is Exercise 3 on Discussion Problems 2.

only holds for k-v.s. since splitting gives us a map $H_i(M) \to Z_i(M) \hookrightarrow M_i$, and this induces the quasi-isom.

For the remainder of the section we assume $(R, \mathfrak{m}, k)$ is local and ω_R is a dualizing complex for R. Note as R-complexes, we have $\mathrm{RHom}_R(k, \omega_R) \simeq \mathrm{Ext}_R(k, \omega_R)$.

1.6.5. Observe that

$$\begin{aligned} k \xrightarrow{\simeq} k^{\dagger\dagger} &\simeq \mathrm{RHom}_R(\mathrm{Ext}_R(k, \omega_R), \omega_R) \\ &\simeq \mathrm{Hom}_k(\mathrm{Ext}_R(k, \omega_R), \mathrm{Ext}_R(k, \omega_R)) \\ &\cong \mathrm{Hom}_k(\mathrm{Ext}_R(k, \omega_R), k) \otimes_k \mathrm{Ext}_R(k, \omega_R)\,. \end{aligned}$$

Therefore, $\mathrm{Ext}_R(k, \omega_R) \simeq \Sigma^a k$ for some integer a. We will say that ω_R is **normalized** if $a = 0$.

1.6.2 (3)

Let ω_R be a normalized dualizing complex for R. Using that $(\omega_R)_\mathfrak{p}$ is a dualizing complex for $R_\mathfrak{p}$, and that *minimal*[6] semiinjective resolutions localize it follows that ω_R admits a ~~minimal~~ minimal semiinjective resolution $\omega_R \xrightarrow{\simeq} I$ where

$$I = 0 \to I_{\dim R} \to I_{\dim R - 1} \to \ldots \to I_1 \to I_0 \to 0$$

where $I_0 = E_R(k)$ and $I_n = \bigoplus_{\dim(R/\mathfrak{p})=n} E_R(R/\mathfrak{p})$. Finally, note that if $\omega_{R_\mathfrak{p}}$ is a normalized dualizing complex for $R_\mathfrak{p}$, then

$$(\omega_R)_\mathfrak{p} \simeq \Sigma^{\dim(R/\mathfrak{p})} \omega_{R_\mathfrak{p}}\,. \tag{1.6.5.1}$$

Theorem 1.6.6. *With the notation above, for any M in $\mathsf{D}^\mathsf{b}(\mathsf{mod}\, R)$ we have:*

(a) $\inf M^\dagger = \mathrm{depth}(M)$;

(b) $\sup M^\dagger = \sup\{\dim R/\mathfrak{p} - \inf M_\mathfrak{p} : \mathfrak{p} \in \mathrm{Spec}\, R\} = \sup\{\dim \mathrm{H}_i(M) - i : i \in \mathbb{Z}\}$.

Proof. By Ischebeck's theorem (see Theorem 1.5.14), we have

$$-\inf M^\dagger = -\inf \mathrm{RHom}_R(M, \omega_R) = \mathrm{inj\,dim}_R(\omega_R) - \mathrm{depth}(M) = -\mathrm{depth}(M)\,;$$

$\sup\{\mathrm{Ext}^i \neq 0\}$

justifying the proof of (a). For (b), observe that by Remark 1.5.15, we have

$$\sup M^\dagger = \sup\{-\mathrm{depth}_{R_\mathfrak{p}}(M^\dagger)_\mathfrak{p} : \mathfrak{p} \in \mathrm{Spec}\, R\}$$

and using (1.6.5.1) we have

$$-\mathrm{depth}_{R_\mathfrak{p}}(M^\dagger)_\mathfrak{p} = \dim(R/\mathfrak{p}) - \mathrm{depth}_{R_\mathfrak{p}}(M_\mathfrak{p})^\dagger\,.$$

$\mathrm{RHom}(M, \omega_R)_\mathfrak{p} \simeq \mathrm{RHom}(M_\mathfrak{p}, (\omega_R)_\mathfrak{p}) \simeq \mathrm{RHom}(M_\mathfrak{p}, \Sigma^{\dim R/\mathfrak{p}} \omega_{R_\mathfrak{p}}) = \Sigma^{\dim R/\mathfrak{p}} (M_\mathfrak{p})^\dagger$

Also, by (a) we have the first equality

$$\mathrm{depth}_{R_\mathfrak{p}}(M_\mathfrak{p})^\dagger = \inf M_\mathfrak{p}^{\dagger\dagger} = \inf M_\mathfrak{p}\,,$$

and the second equality is from Theorem 1.6.4, finishing the proof of (b).[7]

□

Recall that a finitely generated R-module M is **Cohen-Macaulay** if $\mathrm{depth}(M) = \dim(M)$; it is **maximal Cohen-Macaulay** if $\mathrm{depth}(M) = \dim(R)$. The **Cohen-Macaulay defect** of M is

$$\mathrm{cmd}(M) = \dim(M) - \mathrm{depth}(M) \geqslant 0\,.$$

The following is an immediate corollary of the previous theorem.

Corollary 1.6.7. *For a finitely generated R-module M, we have* $\mathrm{cmd}(M) = \sup M^\dagger - \inf M^\dagger$.

Note $\dim 0 = -\infty$

[6] Minimality of a semiinjective R-complex I means that $\mathrm{Hom}_R(k, I)$ has trivial differential. Every R-complex admits a minimal semiinjective resolution. We will not need this explicitly in what follows; however, implicitly, we have used this above to obtain a chain-level "description" of ω_R.

[7] The last equality in (b) is a straightforward calculation and is left to the reader.

1.6.8. Assume $\mathsf{cm}_d(R)$ denote the full subcategory of $\mathsf{mod}(R)$ consisting of Cohen-Macaulay R-modules of dimension d. Given M in $\mathsf{cm}_d(R)$, then $M^\dagger \simeq \Sigma^d \operatorname{H}_d(M^\dagger)$ by the corollary, and the theorem above yields that $\operatorname{H}_d(M^\dagger)$ is again a Cohen-Macaulay module of dimension d. Therefore, we obtain the following.

Corollary 1.6.9. *The self-inverse equivalence $\Sigma^{-d}(-)^\dagger$ on $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ restricts to a bijection from $\mathsf{cm}_d(R)$ to itself.*

Remark 1.6.10. For uniqueness of dualizing complexes, assume ω and ω' are normalized dualizing complexes. We claim that $\omega \simeq P \otimes_R \omega'$ with P a rank one projective. Indeed, it suffices to show if R is local then $\omega \simeq \omega'$. In what follows we use the evaluation isomorphism, from 1.5.13, several times without particular comment.

Note that $\operatorname{RHom}_R(\omega, \omega')$ is in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$, and observe that

$$k \otimes^{\mathrm{L}}_R \operatorname{RHom}_R(\omega, \omega') \simeq \operatorname{RHom}_R(\operatorname{RHom}_R(k, \omega), \omega') \simeq k\,,$$

and so these facts imply that $\operatorname{RHom}_R(\omega, \omega') \simeq R$. Therefore,

$$\omega \simeq \omega \otimes^{\mathrm{L}}_R \operatorname{RHom}_R(\omega, \omega') \simeq \operatorname{RHom}_R(\operatorname{RHom}_R(\omega, \omega), \omega') \simeq \operatorname{RHom}_R(R, \omega') \simeq \omega'\,.$$

1.6.11. Fix M in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$. Recall that its Poincaré series is

$$\mathrm{P}^R_M(t) = \sum_{n \in \mathbb{Z}} \operatorname{rank}_k \operatorname{Tor}^R_n(M, k) t^n = \sum_{n \in \mathbb{Z}} \operatorname{rank}_k \operatorname{Ext}^n_R(M, k) t^n$$

note that $\operatorname{rank}_k \operatorname{Tor}^R_n(M, k)$ is the rank of the nth free module in a *minimal* semifree resolution of M over R. Here minimal is the usual notion in that sense that F is minimal if $\partial(F) \subseteq \mathfrak{m}F$.

The Bass series of M is

$$\mathrm{I}^M_R(t) = \sum_{n \in \mathbb{Z}} \operatorname{rank}_k \operatorname{Ext}^n_R(k, M) t^n$$

using the structure theorem on injectives it follows that $\operatorname{rank}_k \operatorname{Ext}^n_R(k, M)$ is the number of copies of $E_R(k)$ in a minimal semiinjective resolution of M.

Proposition 1.6.12. *For M in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ one has $\mathrm{P}^R_M(t) = \mathrm{I}^{M^\dagger}_R(t)$.*

Proof. Observe that

$$\operatorname{RHom}_R(k, M^\dagger) \simeq \operatorname{RHom}_R(k \otimes^{\mathrm{L}}_R M, \omega) \simeq \operatorname{RHom}_R(M, k^\dagger) \simeq \operatorname{RHom}_R(M, k)\,;$$

the first and second isomorphisms are simply adjunction and the last uses 1.6.5, as ω is a normalized dualizing complex. □

Theorem 1.6.13 (Local duality)**.** *Let $(R, \mathfrak{m}, k)$ be a local ring with normalized dualizing complex ω_R. Set $(-)^\dagger = \operatorname{RHom}_R(-, \omega_R)$ and $(-)^\vee = \operatorname{Hom}_R(-, E_R(k))$. For each M in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$, there is an isomorphism*

$$\mathrm{R}\Gamma_{\mathfrak{m}}(M) \xrightarrow{\simeq} (M^\dagger)^\vee\,.$$

That is, $\operatorname{H}^i_{\mathfrak{m}}(M) \cong \operatorname{H}_i(M^\dagger)^\vee = \operatorname{Hom}_R(\operatorname{Ext}^{-i}_R(M, \omega_R), E_R(k))$ for each $i \in \mathbb{Z}$.

Proof. This is almost an immediate consequence of Grothendieck duality. Namely, Theorem 1.6.4 provides the isomorphism $M \xrightarrow{\simeq} M^{\dagger\dagger}$ and so applying $\mathrm{R}\Gamma_{\mathfrak{m}}$ we obtain the first isomorphism below:

$$\mathrm{R}\Gamma_{\mathfrak{m}}(M) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathfrak{m}}(M^{\dagger\dagger}) = \mathrm{R}\Gamma_{\mathfrak{m}}(\operatorname{RHom}_R(M^\dagger, \omega_R)) \simeq \operatorname{RHom}_R(M^\dagger, \mathrm{R}\Gamma_{\mathfrak{m}}(\omega_R)) \simeq \operatorname{RHom}_R(M^\dagger, E_R(k))\,;$$

the second isomorphism uses Proposition 1.4.15(b) and the last isomorphism follows from the description of ω_R in 1.6.5. □

Part (b) of the next corollary is known as Grothendieck's non-vanishing theorem.

Corollary 1.6.14. (a) $\mathrm{H}^i_{\mathfrak{m}}(M)$ *is artinian* R*-module for each* $i \in \mathbb{Z}$.

(b) *If* M *is a finitely generated* R*-module, then* $-\inf \mathrm{R}\Gamma_{\mathfrak{m}}(M) = \sup\{i \in \mathbb{Z} : \mathrm{H}^i_{\mathfrak{m}}(M) \neq 0\} = \dim(M)$. *That is to say,* $\mathrm{H}^{\dim(M)}_{\mathfrak{m}}(M) \neq 0$ *and* $\mathrm{H}^i_{\mathfrak{m}}(M) = 0$ *for all* $i > \dim(M)$.

Proof. Part (a) from local duality upon recalling Matlis duality Theorem 1.2.7. For part (b), observe that

$$\dim(M) = \sup M^\dagger = -\inf(M^\dagger)^\vee = -\inf \mathrm{R}\Gamma_{\mathfrak{m}}(M)\,;$$

the first equality uses Theorem 1.6.6, the second uses that E is a cogenerator (cf. Lemma 1.2.5), and the last equality is from local duality. □

Remark 1.6.15. When ω is a dualizing complex with $\mathrm{RHom}_R(k, R) \simeq \Sigma^{-a}k$, for some a, then the proof of local duality gives us that

$$\mathrm{H}^i_{\mathfrak{m}}(M) \cong \mathrm{Hom}_R(\mathrm{Ext}^{a-i}_R(M, \omega), E_R(k))$$

for each M in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ and $i \in \mathbb{Z}$. Taking the Matlis dual of both sides also gives

$$\mathrm{Hom}_R(\mathrm{H}^i_{\mathfrak{m}}(M), E_R(k)) \cong \mathrm{Ext}^{a-i}_R(M, \omega)\,.$$

Chapter 2

Gorenstein rings and Maximal Cohen-Macaulay modules

Throughout this chapter we let R be a commutative noetherian ring.

2.1 Gorenstein rings

Definition 2.1.1. We say R is **Gorenstein** if $\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) < \infty$ for each $\mathfrak{p} \in \operatorname{Spec}(R)$. We say R is **strongly Gorenstein** if $\operatorname{inj\,dim}_R(R) < \infty$.

Remark 2.1.2. As finite injective dimension is a property that localizes, it follows that strongly Gorenstein rings are Gorenstein. Note that R is strongly Gorenstein if and only if R is a dualizing complex of itself; in contrast, a strongly Gorenstein ring is exactly a ring that is locally a dualizing complex.

Remark 2.1.3. The ring R is strongly Gorenstein if and only if there exists $d > 0$ such that $\operatorname{Ext}^i_R(M, R) = 0$ for all $i > d$ and M in $\mathsf{mod}(R)$; the d can be taken to be $\operatorname{inj\,dim}_R(R)$. In fact, if R is strongly Gorenstein

$$\operatorname{inj\,dim}_R(R) = \sup\{\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec}(R)\} = \sup\{\operatorname{depth}(R_\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec}(R)\}\,;$$

the second equality uses Theorem 1.5.14. Furthermore, using 1.6.5, it follows that

$$\operatorname{RHom}_{R_\mathfrak{p}}(\kappa(\mathfrak{p}), R_\mathfrak{p}) \simeq \Sigma^{-\dim(R_\mathfrak{p})}\kappa(\mathfrak{p})\,,$$

and so combining this with the fact $\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) = -\inf \operatorname{RHom}_{R_\mathfrak{p}}(\kappa(\mathfrak{p}), R_\mathfrak{p})$, it follows that

$$\operatorname{inj\,dim}_R(R) = \dim(R).$$

Proposition 2.1.4. *A ring R is strongly Gorenstein if and only if it has finite Krull dimension and $\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) < \infty$ for all $\mathfrak{p} \in \operatorname{Spec}(R)$ (equivalently, for all maximal ideal $\mathfrak{p}$ of R). For such rings, $\operatorname{inj\,dim}_R(R) = \dim(R)$.*

Proof. The forward implication was sketched in Remark 2.1.3. For the converse note that

$$\operatorname{inj\,dim}_R(R) = \sup\{\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec}(R)\} = \sup\{\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) : \mathfrak{p} \text{ is a maximal ideal of } R\}\,,$$

and now using that $\operatorname{inj\,dim}_{R_\mathfrak{p}}(R_\mathfrak{p}) = \dim(R_\mathfrak{p})$, as noted in Remark 2.1.3, and

$$\dim(R) = \sup\{\dim(R_\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec}(R)\} = \sup\{\dim(R_\mathfrak{p}) : \mathfrak{p} \text{ is a maximal ideal of } R\}\,,$$

we conclude the desired result. □

Remark 2.1.5. As local rings have finite Krull dimension, being Gorenstein is equivalent to being strongly Gorenstein. The argument in Proposition 2.1.4 shows that a Gorenstein local ring R is Cohen-Macaulay. That is to say, $\operatorname{depth}(R) = \dim(R)$.

Proposition 2.1.6. *Given a regular element x in a local ring $(R, \mathfrak{m}, k)$, then R is Gorenstein if and only if $R/(x)$ is Gorenstein.*

Proof. It follows from the isomorphisms below

$$\begin{aligned}\operatorname{RHom}_R(k, R) &\simeq \operatorname{RHom}_R(k \otimes^{\mathrm{L}}_{R/(x)} R/(x), R)\\ &\simeq \operatorname{RHom}_{R/(x)}(k, \operatorname{RHom}_R(R/(x), R))\\ &\simeq \Sigma^{-1} \operatorname{RHom}_{R/(x)}(k, R/(x))\,;\end{aligned}$$

the isomorphisms are all obvious except the last one, which is where we used that x is a regular element. □

Example 2.1.7. By Example 1.6.2(b), it follows that a regular ring of finite Krull dimension is a strongly Gorenstein ring; in particular, all regular local rings are (strongly) Gorenstein. However, Nagata's example of a noetherian ring with infinite Krull dimension that is locally regular gives an example of a Gorenstein ring that is not strongly Gorenstein.

Example 2.1.8. A local ring $(R, \mathfrak{m})$ is Gorenstein if and only if its $\mathfrak{m}$-adic completion $\widehat{R}$ is Gorenstein. Indeed, the natural map

$$\operatorname{RHom}_R(k, R) \xrightarrow{\simeq} \operatorname{RHom}_R(k, R) \otimes_R \widehat{R} \simeq \operatorname{RHom}_{\widehat{R}}(k, \widehat{R})$$

is an isomorphism. Combining this with Example 2.1.7, as well as Proposition 2.1.6, we have that a complete intersection ring is Gorenstein; recall that such a local ring R is one where its completion $\widehat{R}$ is a regular local ring modulo a regular sequence.

Recall that the type of a local $(R, \mathfrak{m}, k)$ is $\operatorname{rank}_k \operatorname{Ext}^{\dim(R)}_R(k, R)$. For an artinian ring, this is the k-vector space dimension of its socle.

Theorem 2.1.9. *A local ring is Gorenstein if and only if it is Cohen-Macaulay and type one.*

Proof. Let $(R, \mathfrak{m}, k)$ be local. For the forward implication, the fact R is Cohen-Macaulay was already noted in Remark 2.1.5, and we also have from Remark 2.1.3 that $\operatorname{RHom}_R(k, R) \simeq \Sigma^{-\dim R} k$, and so R has type one.

For the converse, we can assume R is complete (cf. Example 2.1.8 and the fact that Cohen-Macaulay and type one is invariant under completion). Hence, by Remark 1.6.3, R has a normalized dualizing complex ω. As R is Cohen-Macualay of dimension, then from local duality we have that $\Sigma^{-d} R^\dagger \simeq \mathrm{H}_d(R^\dagger) = \mathrm{H}_d(\omega)$ is as well; cf. 1.6.8. Also, by Proposition 1.6.12, we have

$$\mu^i_R(R) = \beta^R_i(R^\dagger) = \beta^R_i(\omega) = \beta^R_{i-d}(\mathrm{H}_d(\omega))\,.$$

Apply this with $i = d$, we obtain that $\mathrm{H}_d(\omega)$ is a cyclic R-module and we have isomorphisms

$$R \xrightarrow{\simeq} \operatorname{RHom}_R(\omega, \omega) \simeq \operatorname{RHom}_R(\mathrm{H}_d(\omega), \mathrm{H}_d(\omega))\,;$$

However, this can only be an isomorphism in degree zero homology if $\mathrm{H}_d(\omega) = R$ and so R is its own dualizing complex as

$$R = \mathrm{H}_d(\omega) \simeq \Sigma^{-d}\omega.$$ □

Example 2.1.10. Let k be a field and consider $R = k[x, y, z]/(x^2 - y^2, x^2 - z^2, xy, xz, yz)$. Then R is an artinian ring, and has socle (x^2), which is one dimensional. Hence, by Theorem 2.1.9 implies that R is Gorenstein.

Remark 2.1.11. A theorem of Roberts shows that if R is type one.

2.2 Maximal Cohen-Macaulay modules

Recall a nonzero finitely generated R-module M over a local ring R is **Maximal Cohen-Macaulay** if $\operatorname{depth}(M) = \dim(R)$. Over a general commutative noetherian ring, a nonzero finitely generated R-module is **Maximal Cohen-Macaulay** if it is locally Maximal Cohen-Macaulay. That is to say, $\operatorname{depth}(M_{\mathfrak{p}}) = \dim(R_{\mathfrak{p}})$ for all $\mathfrak{p} \in \operatorname{Spec} R$. We will abbreviate *Maximal Cohen-Macaulay* to MCM. By convention, the zero module is MCM.

Example 2.2.1. (1) A local ring R is MCM as an R-module if and only if it is a Cohen-Macaulay ring.

(2) When R is artinian, then any finitely generated R-module is MCM.

(3) Let $R = k[\![x, y]\!]/(x^2, xy)$ with k a field. Then R is not Cohen-Macaulay. However, the R-module $M = R/(x)$ has depth 1 and hence is MCM.

(4) When R is regular, an R-module is MCM if and only if it is projective. Indeed, a projective module is locally free and regular local rings are Cohen-Macaulay. For the converse, let $\mathfrak{p} \in \operatorname{Spec} R$ and assume $M_{\mathfrak{p}} \neq 0$. T he Auslander-Buchsbaum forumla (Theorem 1.5.9) yields the desired result as:

$$\operatorname{projdim}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \operatorname{depth}(R_{\mathfrak{p}}) - \operatorname{depth}(M_{\mathfrak{p}}) = 0\,.$$

(5) Let $R = k[\![x, y, z, w]\!]/(xw - yz)$ with k a field, and set $M = \operatorname{coker} A$ where

$$A = \begin{bmatrix} x & y \\ z & w \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} w & -y \\ -z & x \end{bmatrix}.$$

Then M is MCM; we will revisit many examples like this later.

Indeed, there is an exact sequence of free R-module

$$\cdots \xrightarrow{B} R^2 \xrightarrow{A} R^2 \xrightarrow{B} R^2 \xrightarrow{A} R^2 \to 0\,.$$

This is a minimal free resolution of M, and a simple depth argument shows that $\operatorname{depth}(M) = 3 = \dim(R)$. Hence, M is MCM.

Proposition 2.2.2. *For a Gorenstein ring R, a finitely generated R-module M is MCM if and only if $\operatorname{Ext}^{\geqslant 1}_R(M, R) = 0$.*

Proof. Can check locally, and now use Ischebeck's formua (cf. Theorem 1.5.14), we obtain

$$\operatorname{depth}(R) - \operatorname{depth}(M) = \sup\{i : \operatorname{Ext}^i_R(M, R) \neq 0\}\,. \qquad \square$$

For the rest of the section, and essentially the remainder of the notes, R will be assumed to be Gorenstein. The following is an easy exercise which is left to the reader.

Proposition 2.2.3. *Given an exact sequence of finitely generated R-modules $0 \to L \to M \to N \to 0$, if N is MCM, then M is MCM if and only if L is.*

The proof of the next classical result is Exercise 2 on Discussion Problems 2.

Lemma 2.2.4 (Schanuel's lemma)**.** *Suppose there are exact sequence $0 \to K_i \to P_i \to M \to 0$ for $i = 1, 2$. If each P_i is a projective R-module, then $K_1 \oplus P_2 \cong K_2 \oplus P_1$.*

Definition 2.2.5. For each $M \in \mathsf{mod}(R)$ choose a surjective R-linear map $P \to M$ with P projective and set $\Omega_R(M) = \operatorname{Ker}(P \to M)$; we call this a **syzygy (module)** of M. Set $\Omega^0_R(M) = M$ and

$$\Omega^n_R(M) = \Omega_R(\Omega^{n-1}_R(M)) \quad \text{for each } n \geqslant 1.$$

When the ring involved is clear, we write $\Omega^n(M)$ for $\Omega^n_R(M)$.

The syzygy modules are only well-defined up to projective summands, but can be made functorial modulo projectives; see Lemma 2.2.4. On the other hand, for $i \geq 1$ the modules $\operatorname{Ext}^i_R(\Omega^n(M), R)$ are independent of this choice. Given this, and Proposition 2.2.3, the following result follows.

Lemma 2.2.6. *Let M be a finitely generated R-module.*

(1) *If $\operatorname{Ext}^{\geqslant d+1}_R(M, R) = 0$ for some integer $d \geqslant 0$, then $\Omega^d(M)$ is MCM.*

(2) *If M is MCM, then so is $\Omega^n(M)$ for each $n \geq 0$.*

In particular, if $\operatorname{inj\,dim} R = d$, then $\Omega^n(M)$ is MCM for $n \geqslant d$. □

In what follows, let $(-)^* = \operatorname{Hom}_R(-, R)$. Also, for each R-module M, there is a natural (in M) biduality map $\delta^M \colon M \to M^{**}$ given by

$$m \mapsto (f \mapsto f(m))$$

for each $f \in M^*$. It is clear that it is an isomorphism when M is a finitely generated projective R-module.

Lemma 2.2.7. *The following statements hold.*

(1) *If $0 \to M \to X_n \to X_{n-1} \to \cdots$ is an acyclic R-complex with each X_i MCM, then M is MCM.*

(2) *If X is an acyclic complex of MCM R-modules, then each $\mathrm{Z}_n(X)$ is MCM and X^* is acyclic.*

Proof. Since a complex is locally acyclic if and only if it is acyclic, and the MCM property is defined locally, we can assume that the ring R is local, and hence that $d := \operatorname{inj\,dim}_R R = \dim R$ is finite; see 2.1.4.

For the proof of (1), set $L = \operatorname{Ker}(X_{n-d+1} \to X_{n-d})$. Consider the exact sequence of R-modules

$$0 \longrightarrow M \longrightarrow X_n \longrightarrow \cdots \longrightarrow X_{n-d+1} \longrightarrow L \longrightarrow 0$$

One has $\operatorname{Ext}^{\geqslant 1}_R(X_i, R) = 0$, as each X_i is MCM; cf. Proposition 2.2.2. Using this a standard exact sequence chase yields the isomorphism below

$$\operatorname{Ext}^j_R(M, R) \cong \operatorname{Ext}^{j+d}_R(L, R) = 0 \quad \text{for } j \geq 1\,;$$

the equality holds because $\operatorname{inj\,dim} R = d$. Therefore, by Lemma 2.2.6, M is MCM.

Next, for (2), it follows from (1) that each $Z_n = \mathrm{Z}_n(X)$ is MCM as there is an exact sequence

$$0 \to Z_n \to X_n \to X_{n-1} \to \cdots$$

with each X_i being MCM. Furthermore, splicing X into short exact sequences $0 \to Z_{n+1} \to X_{n+1} \to Z_n \to 0$, since Z_n is MCM, Proposition 2.2.2 implies that the induced sequence is also exact:

$$0 \longrightarrow (Z_n)^* \longrightarrow (X_{n+1})^* \longrightarrow (Z_{n+1})^* \longrightarrow 0$$

is exact. Given this it follows that X^* is acyclic. □

Theorem 2.2.8. *Let R be a commutative noetherian Gorenstein ring and M a finitely generated R-module. The following conditions are equivalent.*

(1) *M is MCM;*

(2) *M^* is MCM and the biduality map $\delta^M \colon M \to M^{**}$ is an isomorphism;*

(3) *M has a projective co-resolution: An exact sequence*

$$0 \longrightarrow M \longrightarrow P_{-1} \longrightarrow P_{-2} \longrightarrow \cdots$$

where each P_i is a finitely generated projective;

(4) *For any $n \geqslant 0$ there exist a finitely generated R-module L with $\Omega^n(L) \cong M$.*

Proof. Let $P \xrightarrow{\simeq} M$ be a projective resolution of M by finitely generated projectives.

(1)⇒(2): Since M is MCM, then by Proposition 2.2.2 the complex $0 \to M^* \to P_0^* \to P_1^* \to \cdots$ is acyclic. Also, each P_i^* is MCM, since it is projetive, and so Lemma 2.2.7(1) yields that M^* is MCM. Given this Lemma 2.2.7(2) implies the dual complex

$$\cdots \longrightarrow P_1^{**} \longrightarrow P_0^{**} \longrightarrow M^{**} \to 0$$

is acyclic. From the naturality of biduality we have a morphism of complexes

$$\begin{array}{ccccccccccc}
\ldots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \delta^{P_2}} & & \downarrow{\scriptstyle \delta^{P_1}} & & \downarrow{\scriptstyle \delta^{P_0}} & & \downarrow{\scriptstyle \delta^{M}} & & \\
\ldots & \longrightarrow & P_2^{**} & \longrightarrow & P_1^{**} & \longrightarrow & P_0^{**} & \longrightarrow & M^{**} & \longrightarrow & 0
\end{array}$$

and as each P_i is a finitely generated projective R-module, δ^{P_i} is an isomorphism for each i and so δ^M is an isomorphism as claimed.

(2)⇒(3): Let $Q \xrightarrow{\simeq} M^*$ be a projective resolution of M^* by finitely generated projectives. Since M^* is MCM, the complex $0 \to M^{**} \to Q_0^* \to Q_1^* \to \cdots$ is acyclic; cf. Proposition 2.2.2. It remains to use that $M \cong M^{**}$ to that this complex is a projective co-resolution of M.

(3)⇒(4): We can assume $n \geqslant 1$. Set $L := \operatorname{Ker}(P_{-n-1} \to P_{-n-2})$ so that one has a acyclic complex

$$0 \longrightarrow M \longrightarrow P_{-1} \longrightarrow \cdots \longrightarrow P_{-n} \longrightarrow L \longrightarrow 0$$

realizing M as an nth syzygy of L.

(4)⇒(1): Localization preserves exact sequences, so for any prime $\mathfrak{p}$ in R one has $\Omega^n_R(L)_\mathfrak{p} \cong \Omega^n_{R_\mathfrak{p}}(L_\mathfrak{p})$, and so we can assume R is local. In particular, $d := \operatorname{inj\,dim} R$ is finite, and hence the hypothesis yields an L such that $\Omega^d(L) \cong M$, so M is MCM, by Lemma 2.2.6. □

Chapter 3

Stabilization of a Frobenius exact category

3.1 Frobenius exact categories

Definition 3.1.1. An **exact category** $(\mathsf{B}, \mathcal{E})$ is a full, extension-closed subcategory B of an abelian category A along with its short exact sequence $\mathcal{E}$. That is, $\mathcal{E}$ consists of all exact sequences in

$$0 \to X \to Y \to Z \to 0$$

such that X, Y, Z are all objects of B. For an exact sequence

$$0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$$

in $\mathcal{E}$, we call f is called an **admissible mono**, denoted $f\colon X \hookrightarrow Y$, and g is called an **admissible epi**, denoted $g\colon Y \twoheadrightarrow Z$.

3.1.2. Let B be an exact category. Given an exact sequence $0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$ and an object W in B, the following induced sequences of abelian groups are exact:

$$0 \longrightarrow \mathrm{Hom}_{\mathsf{B}}(W, X) \longrightarrow \mathrm{Hom}_{\mathsf{B}}(W, Y) \longrightarrow \mathrm{Hom}_{\mathsf{B}}(W, Z)$$
$$0 \longrightarrow \mathrm{Hom}_{\mathsf{B}}(Z, W) \longrightarrow \mathrm{Hom}_{\mathsf{B}}(Y, W) \longrightarrow \mathrm{Hom}_{\mathsf{B}}(X, W)\,.$$

Example 3.1.3. Every abelian category is exact.

Example 3.1.4. For a commutative noetherian ring R, recall that $\operatorname{mod} R$ is an abelian category. It is clear that $\mathsf{proj}(R)$ and $\mathsf{fl}(R)$ are all exact categories, where the ambient abelian category can be taken to be $\mathsf{mod}(R)$ (or even $\mathsf{Mod}(R)$); in fact, $\mathsf{fl}(R)$ is abelian in its own right while $\mathsf{proj}(R)$ is not.

Definition 3.1.5. Let $(\mathsf{B}, \mathcal{E})$ be an exact category.

(1) An object P of B is **projective** if it satisfies any of the following equivalent conditions:

   (a) $\mathrm{Hom}_{\mathsf{B}}(P, -)$ preserves exact sequences;

   (b) $\mathrm{Hom}_{\mathsf{B}}(P, g)$ is surjective for every admissible epi g;

   (c) every exact sequence $0 \to X \to Y \to P \to 0$ in $\mathcal{E}$ splits.

   We let $\mathsf{proj}(\mathsf{B})$ be the full subcategory of B consisting of the projective objects of B.

(2) We say B has **enough projectives** if for each X in B, there exists an admissible epi $P \twoheadrightarrow X$ where P is projective.

(3) An object I of B is **injective** if it satisfies any of the following equivalent conditions:

   (1) $\mathrm{Hom}_{\mathsf{B}}(-, I)$ preserves exact sequences;

   (2) $\mathrm{Hom}_{\mathsf{B}}(f, I)$ is surjective for every admissible mono f;

   (3) every exact sequence $0 \to I \to Y \to Z \to 0$ in $\mathcal{E}$ splits.

   We let $\mathsf{inj}(\mathsf{B})$ be the full subcategory of B consisting of the injective objects of B.

(4) We say B has **enough injective** if for each X in B, there exists an admissible mono $X \hookrightarrow I$ where I is injective.

Remark 3.1.6. For an exact category B, it is easy to see that $\mathsf{proj}(\mathsf{B})$ and $\mathsf{inj}(\mathsf{B})$ are closed under summands.

Example 3.1.7. Let R be noetherian and consider the abelian category $\mathsf{mod}(R)$, with the usual exact structure (namely, taking all short exact sequences). With this exact structure, $\mathsf{proj}(\mathsf{mod}(R))$ is exactly the category of finitely generated projective R-modules $\mathsf{proj}(R)$. However, if one considers $\mathsf{mod}(R)$, with the exact structure coming only the split exact sequences, then $\mathsf{proj}(\mathsf{mod}(R)) = \mathsf{mod}(R)$.

Example 3.1.8. Consider $\mathsf{mod}(R)$ with the standard exact structure (coming from the fact it is abelian). For a commutative artinian Gorenstein ring R,

$$\mathsf{proj}(\mathsf{mod}(R)) = \mathsf{proj}(R) = \mathsf{inj}(\mathsf{mod}(R))\,.$$

When R has positive Krull dimension, then

$$\mathsf{proj}(\mathsf{mod}(R)) = \mathsf{proj}(R) \quad \text{and} \quad \mathsf{inj}(\mathsf{mod}(R)) = 0.$$

Definition 3.1.9. Let B be an exact category. We say that B is **Frobenius** if it has enough injectives, enough projectives and $\mathsf{proj}(\mathsf{B}) = \mathsf{inj}(\mathsf{B})$.

3.1.10. For a commutative noetherian Gorenstein ring R, write $\mathsf{MCM}(R)$ full the subcategory of $\mathsf{mod}(R)$ of MCM R-modules. Note by Theorem 2.2.8, it follows that this category is closed under R-linear duality $(-)^* = \mathrm{Hom}_R(-, R)$. Furthermore, the biduality maps $\delta^M \colon M \to M^{**}$ are isomorphisms for all M in $\mathsf{MCM}(R)$.

Proposition 3.1.11. *For a commutative noetherian Gorenstein ring R, its category of MCM R-modules $\mathsf{MCM}(R)$ is Frobenius exact and $\mathsf{proj}(\mathsf{MCM}(R)) = \mathsf{proj}(R)$.*

Proof. From Proposition 2.2.3 it follows that $\mathsf{MCM}(R)$ is extension-closed in $\mathsf{mod}(R)$ and so it is, indeed, an exact category. Furthermore, from 3.1.10 we have that $(-)^*$ is a self-inverse equivalence below:

$$\begin{array}{ccc} \mathsf{MCM}(R) & \xrightarrow[\equiv]{(-)^*} & \mathsf{MCM}(R) \\ {\scriptstyle\subseteq}\uparrow & & \uparrow{\scriptstyle\subseteq} \\ \mathsf{proj}(\mathsf{MCM}(R)) & \xrightarrow[\equiv]{(-)^*} & \mathsf{inj}(\mathsf{MCM}(R))\,. \end{array}$$

Indeed, the top level equivalence is directly in 3.1.10. For restriction, if M is projective in $\mathsf{MCM}(R)$, then observe that

$$\mathrm{Hom}_R(-, M^*) \cong \mathrm{Hom}_R(M, (-)^*)$$

and so M^* is injective in $\mathsf{MCM}(R)$. Also, if M is injective in $\mathsf{MCM}(R)$, then we have the following isomorphisms on $\mathsf{MCM}(R)$:

$$\mathrm{Hom}_R(M^*, -) \cong \mathrm{Hom}_R(M^*, (-)^{**}) \cong \mathrm{Hom}_R((-)^*, M^{**}) \cong \mathrm{Hom}_R((-)^*, M)\,.$$

Hence, M^* is projective in $\mathsf{MCM}(R)$.

Next, we claim that $\mathsf{proj}(\mathsf{MCM}(R)) = \mathsf{proj}(R)$. It is clear that each objects of $\mathsf{proj}(R)$ is an object of $\mathsf{proj}(\mathsf{MCM}(R))$. Conversely, given an object M of $\mathsf{proj}(\mathsf{MCM}(R))$, consider the exact sequence (in $\mathsf{mod}(R)$):

$$0 \to K \to P \to M \to 0$$

where P is in $\mathsf{proj}(R)$. Again, Proposition 2.2.3 says that K is MCM and so this is an exact sequence of MCM modules. Since M is in $\mathsf{proj}(\mathsf{MCM}(R))$, the exact sequence splits and so M is a summand of P; hence, M is in $\mathsf{proj}(R)$. Furthermore, it is now clear that $\mathsf{MCM}(R)$ has enough projectives.

From the duality square above, since $\mathsf{proj}(\mathsf{MCM}(R)) = \mathsf{proj}(R)$ it follows that $\mathsf{inj}(\mathsf{MCM}(R))$. To show there are enough injective, let M be MCM and consider an epi $P \to M^*$, dualizing and again making use of 3.1.10, we obtain an admissible mono

$$M \xrightarrow{\cong} M^{**} \hookrightarrow P^*$$

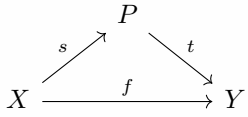
and since P^* is injective in $\mathsf{MCM}(R)$, we have verified there are enough injectives, completing the proof. □

Example 3.1.12. Let A be an abelian category. The category of complexes over A forms a Frobenius exact category; the exact structure comes from the short exact sequence of complexes that are split exact degreewise. A projective/injective object is a summand of a cone on the identity map of a complex over A.[1]

3.2 Stablization

Throughout this section $(\mathsf{B}, \mathcal{E})$ will be a Frobenius (exact) category.

Definition 3.2.1. A map $f\colon X \to Y$ is **phantom** if it factors through a projective, that is to say, there exists a commutative diagram



with P projective. The subset of phantom maps from X to Y is denoted $\mathrm{PHom}_{\mathsf{B}}(X, Y)$. This is a subgroup of $\mathrm{Hom}_{\mathsf{B}}(X, Y)$, since $\mathsf{proj}(\mathsf{B})$ is additive.

Lemma 3.2.2. *The phantom maps form an ideal in* B.

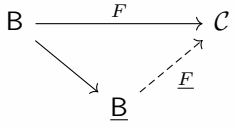
Proof. An ideal in B is a collection of maps that is closed under addition (when possible) and compositions (on either side) when sensible. It is clear from the definition that the phantom maps form an ideal in B. □

Definition 3.2.3. The **stable category** of $(\mathsf{B}, \mathcal{E})$, denoted $\underline{\mathsf{B}}$, is the quotient category $\mathsf{B}/\,\mathrm{PHom}_{\mathsf{B}}(-,-)$. Thus objects of $\underline{\mathsf{B}}$ are the objects of B, and the morphisms between X, Y in $\underline{\mathsf{B}}$ are given by

$$\underline{\mathrm{Hom}}_{\mathsf{B}}(X, Y) := \frac{\mathrm{Hom}_{\mathsf{B}}(X, Y)}{\mathrm{PHom}_{\mathsf{B}}(X, Y)}$$

It is not hard to verify that this is an additive category, and that the canonical map $\mathsf{B} \to \underline{\mathsf{B}}$, which is the identity on objects, is an additive, full functor.

Remark 3.2.4. The construction of the stable category solves the following universal problem: Any additive functor $F\colon \mathsf{B} \to \mathcal{C}$ to any additive category $\mathcal{C}$ such that $F(f) = 0$ for any phantom map f factors uniquely through $\underline{\mathsf{B}}$:



[1] This is Exercise 2 on Discussion Problems 3.

3.2.5. An object X is 0 in $\underline{\mathsf{B}}$ if and only if X is in $\mathsf{proj}(\mathsf{B})$. Indeed, $X \cong 0$ in $\underline{\mathsf{B}}$ if and only if $\mathrm{id}_X = 0$ in $\underline{\mathsf{B}}$. That is, there exists a factorization

$$\begin{array}{ccc} & P & \\ \nearrow & & \searrow \\ X & \xrightarrow{\mathrm{id}_X} & X \end{array}$$

where P is in $\mathsf{proj}(\mathsf{B})$. This is clearly equivalent to X itself being in $\mathsf{proj}(\mathsf{B})$ (see 3.1.6).

Lemma 3.2.6. *For any object X in B and projective P, the canonical monomorphism $X \rightarrowtail X \oplus P$ is an isomorphism in $\underline{\mathsf{B}}$, with inverse the canonical epimorphism $X \oplus P \to X$.*

Proof. Let i and p be the monomorphism and epimorphism in question. Evidently $pi = \mathrm{id}_X$, whilst the map $\mathrm{id}_{X\oplus P} - ip$ factors as

$$\begin{array}{ccc} & P & \\ {\scriptstyle s}\nearrow & & \searrow{\scriptstyle t} \\ X \oplus P & \xrightarrow[\mathrm{id}_{X\oplus P} - ip]{} & X \oplus P \end{array}$$

where s and t are the canonical maps. Thus $\mathrm{id}_{X\oplus P} = ip$ in $\underline{\mathsf{B}}$. □

Proposition 3.2.7. *A map $f\colon X \to Y$ is an isomorphism in $\underline{\mathsf{B}}$ if and only if there is commutative diagram*

$$\begin{array}{ccc} X \oplus P & \xrightarrow[\cong]{\widetilde{f}} & Y \oplus Q \\ \uparrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

in B where P and Q are projective and the vertical maps are the canonical ones.

Proof. Given Lemma 3.2.6, the existence of a diagram as in the statement readily implies that f is an isomorphism in $\underline{\mathsf{B}}$. The proof of the converse is a bit more elaborate, and goes through a series of reductions.

The hypothesis is that there is a map $g\colon Y \to X$ with $gf = \mathrm{id}_X$ and $fg = \mathrm{id}_Y$ in $\underline{\mathsf{B}}$.

Claim (1). One can assume $gf = \mathrm{id}_X$ in B.

Indeed $gf = \mathrm{id}_X$ in $\underline{\mathsf{B}}$ means that $\mathrm{id}_X - gf$ factors as $X \xrightarrow{s} P \xrightarrow{t} X$ where P is a projective. Evidently the square in the following diagram commutes:

$$\begin{array}{ccccc} X & \xrightarrow{\begin{bmatrix} s \\ f \end{bmatrix}} & P \oplus Y & \xrightarrow{[t\ g]} & X \\ \| & & \downarrow{\scriptstyle [0\ 1]} & & \\ X & \xrightarrow{f} & Y & & \end{array}$$

Since the epimorphism above is an isomorphism in $\underline{\mathsf{B}}$, so is $[s\ f]^T$. The composition of the maps in the top row is id_X, so replacing f and g by $[s\ f]^T$ and $[t\ g]$, respectively, justifies the claim.

Claim (2). One can assume the monomorphism f is admissible.

Indeed, since B has enough injectives, and projectives are injectives, there is an admissible monomorphism $i\colon X \hookrightarrow P$ with P projective. The push-out of i along f yields a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{i} & P & \longrightarrow & X' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow & & \| & & \\ 0 & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & X' & \longrightarrow & 0 \end{array}$$

Since B is extension-closed it follows that Z is in B. Consider the following diagram where the first row is the one that defines the push-out

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{[-i\ f]^T} & P \oplus Y & \longrightarrow & Z & \longrightarrow & 0 \\ & & \| & & \downarrow & & & & \\ 0 & \longrightarrow & X & \xrightarrow{f} & Y & & & & \end{array}$$

Observe that the composition $P \oplus Y \twoheadrightarrow P \xrightarrow{g} X$ is a left-inverse of $[-i\ f]^T$ and that $[-i\ f]^T \circ g$ is the identity on $P \oplus Y$ in $\underline{\mathsf{B}}$; this again uses Lemma 3.2.6. Replacing f by $[-i\ f]^T$ justifies the claim.

At this point one has a split-exact sequence

$$0 \longrightarrow X \underset{f}{\overset{g}{\leftrightarrows}} Y \underset{t}{\overset{s}{\leftrightarrows}} Z \longrightarrow 0$$

where the map s is induced by $\mathrm{id}_Y - fg$, and satisfies $ts = \mathrm{id}_Z$. This yields the isomorphism in the top row of the commutative diagram

$$\begin{array}{ccc} X \oplus Z & \xrightarrow[\cong]{[f\ s]^T} & Y \\ \uparrow & & \| \\ X & \xrightarrow{f} & Y \end{array}$$

It remains to verify the following:

Claim (3). Z is projective.

Indeed, now is when we use the hypothesis that $fg = \mathrm{id}_Y$ in $\underline{\mathsf{B}}$. This means that $fg - \mathrm{id}_Y$ is a phantom map in B, and hence so is $t(fg - \mathrm{id}_Y)s$. It remains to note that this latter map equals id_Z; this is part of the splitting data.

This completes the proof of the proposition. □

3.3 (Co-)Syzygy functors

Let $(\mathsf{B}, \mathcal{E})$ be a Frobenius exact category and $\underline{\mathsf{B}}$ its stabilization. Next we verify that the construction of syzygies and cosyzygies can be made functorially on $\underline{\mathsf{B}}$. The fundamental result is the following.

Lemma 3.3.1. *Consider exact sequences in the rows of the following diagram, where P and Q are projective, and a map $f\colon X \to Y$.*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X' & \xrightarrow{s_X} & P & \xrightarrow{p_X} & X & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle \widetilde{f}} & & \downarrow{\scriptstyle f} & & \\ 0 & \longrightarrow & Y' & \xrightarrow{s_Y} & Q & \xrightarrow{p_Y} & Y & \longrightarrow & 0 \end{array}$$

There exist maps $\widetilde{f}$ and f' as above making the diagram commute. Moreover, the residue class of the map f' in $\underline{\mathsf{B}}$ is independent of the choice of $\widetilde{f}$.

Proof. As P is projective there is a map $\widetilde{f}\colon P \to Q$ such that $p_Y \widetilde{f} = f p_X$, and this induces f'. For uniqueness of the lift, suppose $\widetilde{g}\colon P \to Q$ and $g'\colon X' \to Y'$ also make the diagram commute. Then $p_Y(\widetilde{f} - \widetilde{g}) = 0$ and hence there exists a map $h\colon P \to Y'$ such that $s_Y h = \widetilde{f} - \widetilde{g}$. Since

$$s_Y(f' - g') = (\widetilde{f} - \widetilde{g})s_X = s_Y h s_X$$

and s_Y is a monomorphism one has $f' - g' = hs_X$. Thus $f' - g'$ factors through a projective, and so is phantom. □

The following is the dual version of the previous lemma; its proof is almost identical and left as an exercise.

Lemma 3.3.2. *Consider exact sequences in the rows of the following diagram, where P and Q are projective, and a map $f\colon X \to Y$.*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{s_X} & P & \xrightarrow{p_X} & X' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \widetilde{f}} & & \downarrow{\scriptstyle f'} & & \\ 0 & \longrightarrow & Y & \xrightarrow{s_Y} & Q & \xrightarrow{p_Y} & Y' & \longrightarrow & 0 \end{array}$$

There exist maps $\widetilde{f}$ and f' as above making the diagram commute. Moreover, the residue class of the map f' in $\underline{\mathsf{B}}$ is independent of the choice of $\widetilde{f}$.

Construction 3.3.3. Invoking the axiom of choice, for each X in B we fix an exact sequence

$$0 \to K_X \to P_X \to X \to 0$$

with P_X an object of $\mathsf{proj}(\mathsf{B})$. We define ΩX to be K_X, called the syzygy of X. By Lemma 3.3.1, this defines an additive functor $\Omega\colon \underline{\mathsf{B}} \to \underline{\mathsf{B}}$; indeed, on objects it is the assignment $X \mapsto \Omega X$ and for each morphism $f\colon X \to Y$ we use Lemma 3.3.1 to construct

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Omega X & \longrightarrow & P_X & \longrightarrow & X & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle \tilde{f}} & & \downarrow{\scriptstyle f} & & \\ 0 & \longrightarrow & \Omega Y & \longrightarrow & P_Y & \longrightarrow & Y & \longrightarrow & 0 \end{array}$$

and assign f to $\Omega f := f'$, which is unique in $\underline{\mathsf{B}}$. Functoriality is clear enough from the unique of Ωf in $\underline{\mathsf{B}}$.

Note that for X in $\mathsf{proj}(\mathsf{B})$ we have that ΩX is in projective. Indeed, the defining exact sequence

$$0 \to \Omega X \to P_X \to X \to 0$$

splits since X is projective. Thus, ΩX is a summand of P_X which is itself projective. Therefore, ΩX is projective (cf. 3.1.6). In particular, for any phantom $f\colon X \to Y$ we factor it as

$$\begin{array}{ccccc} & & P & & \\ & \nearrow & & \searrow & \\ X & & \xrightarrow{\quad f \quad} & & Y \end{array}$$

with P projective and so applying Ω shows that $\Omega f = 0$. Thus, the universal property of $\underline{\mathsf{B}}$, described in 3.2.4, defines a functor $\Omega\colon \underline{\mathsf{B}} \to \underline{\mathsf{B}}$ that fits into the diagram

$$\begin{array}{ccc} \mathsf{B} & \xrightarrow{\Omega} & \underline{\mathsf{B}} \\ \downarrow & \nearrow{\scriptstyle \Omega} & \\ \underline{\mathsf{B}} & & \end{array}$$

where the vertical map is the canonical stabilization functor.

Definition 3.3.4. The functor $\Omega : \underline{\mathsf{B}} \to \underline{\mathsf{B}}$ from Construction 3.3.3 is called the *syzygy functor on* $\underline{\mathsf{B}}$.

Remark 3.3.5. It is worth remarking that Ω is well-defined up to natural isomorphism. Indeed, suppose Ω' was constructed using a different choice of exact sequences (again, invoking the axiom of choice)

$$0 \to \Omega' X \to P'_X \to X \to 0$$

for each X in $\underline{\mathsf{B}}$ where each P'_X is projective. For each X, two applications of Lemma 3.3.1 define maps in the commutative diagram below:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \Omega X & \longrightarrow & P_X & \longrightarrow & X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \eta_X} & & \downarrow{\scriptstyle \tilde{f}} & & \downarrow{\scriptstyle =} & & \\
0 & \longrightarrow & \Omega' X & \longrightarrow & P'_X & \longrightarrow & X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \eta'_X} & & \downarrow{\scriptstyle \tilde{g}} & & \downarrow{\scriptstyle =} & & \\
0 & \longrightarrow & \Omega X & \longrightarrow & P_X & \longrightarrow & XX & \longrightarrow & 0
\end{array}$$

and also imply that $\eta'_X \eta_X = \mathrm{id}_{\Omega X}$ and $\eta_X \eta'_X = \mathrm{id}_{\Omega' X}$. It remains to check the naturality of η. This follows easily by noticing that for any $f\colon X \to Y$, both $\Omega' f \eta_X$ and $\eta'_y \Omega f$ are lifts of f to $\Omega X \to \Omega' Y$ and one last application of 3.3.1 shows these must, in fact, be equal. Thus, Ω and Ω' are naturally isomorphic via η.

Construction 3.3.6. Here we construct a cosyzygy functor on $\underline{\mathsf{B}}$. The construction is dual so we move more briskly through this construction.

Invoking the axiom of choice, for each X in B we fix an exact sequence

$$0 \to X \to P^X \to C^X \to 0$$

with P^X an object of $\mathsf{proj}(\mathsf{B})$. We define $\Omega^{-1} X$ to be C_X, called the *cosyzygy of* X. By Lemma 3.3.2, this defines an additive functor a functor $\Omega^{-1}\colon \underline{\mathsf{B}} \to \underline{\mathsf{B}}$ that fits into the diagram

$$\begin{array}{ccc}
\mathsf{B} & \xrightarrow{\Omega^{-1}} & \underline{\mathsf{B}} \\
\downarrow & \nearrow_{\Omega^{-1}} & \\
\underline{\mathsf{B}} & &
\end{array}$$

where the vertical map is the canonical stabilization functor. Any other choice of exact sequences defining the objects $\Omega^{-1} X$ yields a naturally isomorphic functor.

Definition 3.3.7. The functor $\Omega^{-1}\colon \underline{\mathsf{B}} \to \underline{\mathsf{B}}$ from Construction 3.3.6 is called the *cosyzygy functor on* $\underline{\mathsf{B}}$.

Theorem 3.3.8. *The functors Ω and Ω^{-1} are quasi-inverse auto-equivalences of $\underline{\mathsf{B}}$. That is, $\Omega\Omega^{-1}$ and $\Omega^{-1}\Omega$ are naturally isomorphic to the identity functor on $\underline{\mathsf{B}}$.*

Proof. We check $\Omega^{-1}\Omega$ is naturally isomorphic to the identity functor on $\underline{\mathsf{B}}$. For each X in $\underline{\mathsf{B}}$, by Lemma 3.3.1 we obtain maps and a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \Omega X & \longrightarrow & P_X & \longrightarrow & X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle =} & & \downarrow{\scriptstyle \tilde{f}} & & \downarrow{\scriptstyle \eta_X} & & \\
0 & \longrightarrow & \Omega X & \longrightarrow & P^{\Omega X} & \longrightarrow & \Omega^{-1}\Omega X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle =} & & \downarrow{\scriptstyle \tilde{g}} & & \downarrow{\scriptstyle \epsilon_X} & & \\
0 & \longrightarrow & \Omega X & \longrightarrow & P_X & \longrightarrow & X & \longrightarrow & 0
\end{array}$$

The same arguments as in Remark 3.3.5, but with the dual Lemma 3.3.2, shows that η_X and ϵ_X are inverse isomorphisms and η is in fact natural. □

3.4 Triangulated structure on the stable category

Throughout this chapter $(\mathsf{B}, \mathcal{E})$ will be a Frobenius exact category and $\underline{\mathsf{B}}$ its stabilization.

Definition 3.4.1. Let $f\colon X \to Y$ be a morphism in B and $i_X\colon X \to P^X$ the morphism from Construction 3.3.6. Consider the commutative diagram obtain by the pushout of f along i_X:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{i_X} & P^X & \longrightarrow & \Omega^{-1}X & \longrightarrow & 0 \\ & & {\scriptstyle f}\downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & Y & \xrightarrow[g]{} & C_f & \xrightarrow[h]{} & \Omega^{-1}X & \longrightarrow & 0 \end{array}$$

The **distinguished triangle** arising from f is the image in $\underline{\mathsf{B}}$ of the diagram

$$X \xrightarrow{f} Y \xrightarrow{g} C_f \xrightarrow{h} \Omega^{-1}(X)\,.$$

A **triangle** in $\underline{\mathsf{B}}$ is any diagram which is isomorphic to a distinguished triangle. That is to say, a diagram $U \to V \to Z \to \Omega^{-1}U$ is a triangle if there is a commutative diagram

$$\begin{array}{ccccccc} U & \longrightarrow & V & \longrightarrow & W & \longrightarrow & \Omega^{-1}U \\ {\scriptstyle u}\downarrow\cong & & \downarrow\cong & & \downarrow\cong & & {\scriptstyle \Omega^{-1}u}\downarrow\cong \\ X & \xrightarrow{f} & Y & \longrightarrow & C_f & \longrightarrow & \Omega^{-1}X \end{array}$$

in $\underline{\mathsf{B}}$ with bottom row a distinguished triangle and vertical maps isomorphisms.

3.4.2. Let $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ be a short exact sequence in B; this determines an exact triangle in $\underline{\mathsf{B}}$. Indeed, consider the commutative diagram in B

$$\begin{array}{ccccccccc} & & 0 & & 0 & & & & \\ & & \downarrow & & \downarrow & & & & \\ 0 & \longrightarrow & X & \xrightarrow{i^X} & P^X & \xrightarrow{\pi^X} & \Omega^{-1}X & \longrightarrow & 0 \\ & & {\scriptstyle f}\downarrow & & \downarrow & & {\scriptstyle =}\downarrow & & \\ 0 & \longrightarrow & Y & \xrightarrow{s} & C_f & \xrightarrow{t} & \Omega^{-1}X & \longrightarrow & 0 \\ & & {\scriptstyle g}\downarrow & & {\scriptstyle q}\downarrow & & & & \\ & & Z & \xrightarrow{=} & Z & & & & \\ & & \downarrow & & \downarrow & & & & \\ & & 0 & & 0 & & & & \end{array}$$

with exact rows; the left vertical sequence is exact by assumption and the middle exact sequence is induced by the Snake Lemma (in the ambient abelian category). The thing to note is that the image of q in $\underline{\mathsf{B}}$ is an isomorphism. Hence, in $\underline{\mathsf{B}}$ we have an exact triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{tq^{-1}} \Omega^{-1}X$$

which is isomorphic to the distinguished one corresponding to f. Conversely, any triangle in $\underline{\mathsf{B}}$ comes from a short exact sequence in this way.

Theorem 3.4.3. *The additive category $\underline{\mathsf{B}}$ with the specified class of triangles in Definition 3.4.1 is a triangulated category, with suspension functor Ω^{-1}.*

Proof. In Theorem 3.3.8, we already showed Ω^{-1} is an auto-equivalence of $\underline{\mathsf{B}}4$, and so set $\Sigma := \Omega^{-1}$ on $\underline{\mathsf{B}}$. Also, the class of exact triangles are closed under isomorphism and each morphism fits into a triangle as discussed above. So it remains to directly check the axioms of a triangulated category to show $(\underline{\mathsf{B}}, \Omega^{-1})$ is in fact triangulated.

(TR1): $0 \to X \xrightarrow{\mathrm{id}_X} X \to 0$ shoud be a triangle for any $X \in \underline{\mathsf{B}}$. Indeed, the pushout along $\begin{array}{ccc} 0 & \xrightarrow{i^0} & P^0 \\ \downarrow & & \\ X & & \end{array}$ gives us the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \xrightarrow{i^0} & P^0 & \longrightarrow & \Omega^{-1}0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow{=} & & \\ 0 & \longrightarrow & X & \longrightarrow & X \oplus P^0 & \longrightarrow & \Omega^{-1}0 & \longrightarrow & 0 \end{array}$$

in B with exact rows. By Lemma 3.2.6, we have an isomorphism of diagrams in $\underline{\mathsf{B}}$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & X & \xrightarrow{\mathrm{id}_X} & X & \longrightarrow & 0 \\ \downarrow{=} & & \downarrow{=} & & \downarrow{\cong} & & \downarrow{\cong} \\ 0 & \longrightarrow & X & \longrightarrow & X \oplus P^0 & \longrightarrow & \Omega^{-1}0\,. \end{array}$$

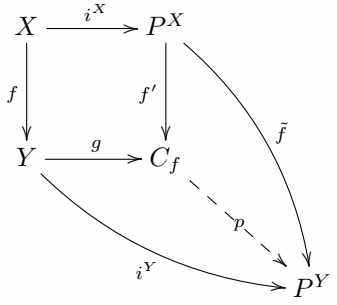
(TR2): Given a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$, then the sequence $Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X \xrightarrow{-\Sigma f} \Sigma Y$ is a triangle. It suffices to check this holds for the distinguished triangles. Let $X \xrightarrow{f} Y \xrightarrow{g} C_f \xrightarrow{h} \Omega^{-1}X$ be a distinguished triangle. Then this is obtained from the commutative diagram in B whose left square is a pushout of f along i^X:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{i^X} & P^X & \xrightarrow{\pi^X} & \Omega^{-1}X & \longrightarrow & 0 \\ & & \downarrow{f} & & \downarrow{f'} & & \downarrow{=} & & \\ 0 & \longrightarrow & Y & \xrightarrow{g} & C_f & \xrightarrow{h} & \Omega^{-1}X & \longrightarrow & 0 \end{array}$$

Now considering the commutative diagram in B, from Lemma 3.3.2,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{i^X} & P^X & \xrightarrow{\pi^X} & \Omega^{-1}X & \longrightarrow & 0 \\ & & \downarrow{f} & & \downarrow{\tilde{f}} & & \downarrow{\Omega^{-1}f} & & \\ 0 & \longrightarrow & Y & \xrightarrow{i^Y} & P^Y & \xrightarrow{\pi^Y} & \Omega^{-1}Y & \longrightarrow & 0 \end{array}$$

there exists a morphism $p\colon C_f \to P^Y$ such that



commutes in B and $\Omega^{-1}fh = \pi^Y p$. Thus, we have a commutative diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & & & \\
 & & \downarrow & & \downarrow & & & & \\
0 & \longrightarrow & Y & \xrightarrow{g} & C_f & \xrightarrow{h} & \Omega^{-1}X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle i^Y} & & \downarrow{\scriptstyle \binom{p}{h}} & & \downarrow{\scriptstyle =} & & \\
0 & \longrightarrow & P^Y & \longrightarrow & P^Y \oplus \Omega^{-1}X & \longrightarrow & \Omega^{-1}X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi^Y} & & \downarrow{\scriptstyle (\pi^Y \quad -\Omega^{-1}f)} & & & & \\
 & & \Omega^{-1}Y & \xrightarrow{=} & \Omega^{-1}Y & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0 & & & &
\end{array}$$

in B with exact rows and columns. From this we obtain a distinguished triangle

$$Y \xrightarrow{g} C_f \xrightarrow{\binom{p}{h}} P^Y \oplus \Omega^{-1}X \xrightarrow{(\pi^Y \quad -\Omega^{-1}f)} \Omega^{-1}Y$$

in $\underline{\mathsf{B}}$ which is isomorphic to $Y \xrightarrow{g} C_f \xrightarrow{h} \Omega^{-1}X \xrightarrow{-\Omega^{-1}f} \Omega^{-1}Y$, again using Lemma 3.2.6.

(TR3): Consider the commutative diagram in $\underline{\mathsf{B}}$

$$\begin{array}{ccccccc}
X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & \Omega^{-1}X \\
\downarrow{\scriptstyle x} & & \downarrow{\scriptstyle y} & & & & \downarrow{\scriptstyle \Sigma x} \\
X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & \Omega^{-1}X'
\end{array}$$

We need to show there exists $z\colon Z \to Z'$ that makes the two right-hand squares commute. We first prove this when $f'x = yf$ in B. Considering the pushout diagrams defining Z and Z',

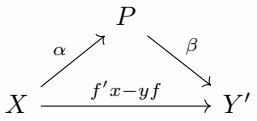
$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \xrightarrow{i^X} & P^X & \xrightarrow{\pi^X} & \Omega^{-1}X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \overline{f}} & & \downarrow{\scriptstyle =} & & \\
0 & \longrightarrow & Y & \xrightarrow{g} & Z & \xrightarrow{h} & \Omega^{-1}X & \longrightarrow & 0
\end{array}$$

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X' & \xrightarrow{i^{X'}} & P^{X'} & \xrightarrow{\pi^{X'}} & \Omega^{-1}X' & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle \overline{f'}} & & \downarrow{\scriptstyle =} & & \\
0 & \longrightarrow & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & \Omega^{-1}X' & \longrightarrow & 0
\end{array}$$

the assumption that $f'x = yf$ in B induces a map $z\colon Z \to Z'$ satisfying

$$g'y = zg \quad \text{and} \quad h'z = \Omega^{-1}xh.$$

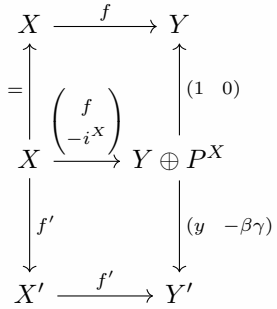
Now assume that $f'x = yf$ in $\underline{\mathsf{B}}$. That is, $f'x - yf$ factors through a projective P as



in B. Since P is also injective, the admissible mono i^X induces a map $\gamma\colon P^X \to P$ such that

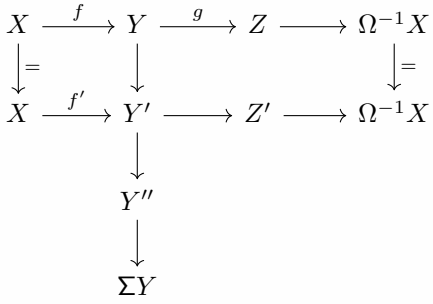
$$\beta\gamma i^X = f'x - yf$$

in B. We use this to construct the following commutative diagram in B



Now the general case follows from the previously handled one.

(TR4): Finally, we need to check the octrahedral axiom holds. Consider triangles with a commutative square given a commutative diagram in $\underline{\mathsf{B}}$



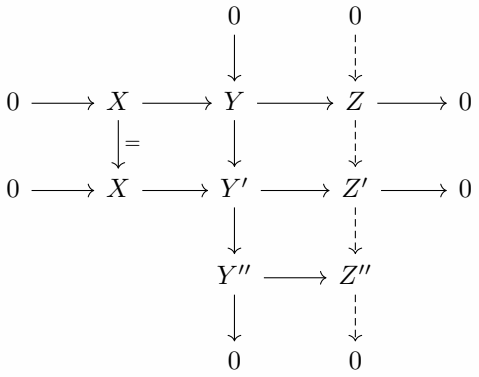
Note that the image of the admissible short exact sequence

$$0 \to X \xrightarrow{\begin{pmatrix} f \\ -i^X \end{pmatrix}} Y \oplus P^X \xrightarrow{(g \quad 0)} Z \to 0$$

in B is isomorphic in $\underline{\mathsf{B}}$ to the exact triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \to \Omega^{-1}X.$$

Therefore, we can reset notation and the diagram above, corresponds to the diagram of short exact sequences in B:



where the rows and columns are short exact sequences in B. Now an application of the Snake Lemma establishes the octahedral axiom for $\underline{\mathsf{B}}$. □

3.4.4. Let $F\colon \mathsf{B} \to \mathsf{C}$ be an exact map of Frobenius exact categories; that is to say, F is additive and preserves exact sequences. If F, in addition, $F(\mathsf{proj}(\mathsf{B})) \subseteq \mathsf{proj}(\mathsf{C})$ and $F(\Omega_{\mathsf{B}}-) \simeq \Omega_{\mathsf{C}}F(-)$, then we obtain an exact functor of triangulated categories $\underline{F}\colon \underline{\mathsf{B}} \to \underline{\mathsf{C}}$ fitting into the commutative diagram

$$\begin{array}{ccc} \mathsf{B} & \xrightarrow{F} & \mathsf{C} \\ \downarrow & & \downarrow \\ \underline{\mathsf{B}} & \xrightarrow[\underline{F}]{} & \underline{\mathsf{C}} \end{array}$$

where the vertical maps are the canonical stabilization functors.

3.4.5. Given a triangulated category (T, Σ), then for any triangle $X \to Y \to Z \to \Sigma X$ and W in T we obtain exact sequences:

$$\cdots \to \mathrm{Hom}_{\mathsf{T}}(W, \Sigma^{i-1}Z) \to \mathrm{Hom}_{\mathsf{T}}(W, \Sigma^{i}X) \to \mathrm{Hom}_{\mathsf{T}}(W, \Sigma^{i}Y) \to \mathrm{Hom}_{\mathsf{T}}(W, \Sigma^{i}Z) \to \mathrm{Hom}_{\mathsf{T}}(W, \Sigma^{i+1}X) \to \cdots$$

$$\cdots \to \mathrm{Hom}_{\mathsf{T}}(\Sigma^{i+1}X, W) \to \mathrm{Hom}_{\mathsf{T}}(\Sigma^{i}Z, W) \to \mathrm{Hom}_{\mathsf{T}}(\Sigma^{i}Y, W) \to \mathrm{Hom}_{\mathsf{T}}(\Sigma^{i}X, W) \to \mathrm{Hom}_{\mathsf{T}}(\Sigma^{i-1}Z, W) \to \cdots$$

Definition 3.4.6. Let R be a commutative noetherian Gorenstein ring, and $\underline{\mathsf{MCM}}(R)$ the stable category of MCM modules. For MCM R-modules M, N, the *Tate cohomology* in degree i is the R-module

$$\widehat{\mathrm{Ext}}^i_R(M,N) := \underline{\mathrm{Hom}}_R(M, \Sigma^i N) = \underline{\mathrm{Hom}}_R(M, \Omega^{-i}(N))\,.$$

Corollary 3.4.7. $(\underline{\mathsf{MCM}}(R), \Omega^{-1})$ *is a triangulated category with triangles defined in Definition 3.4.1.*

Remark 3.4.8. Given an exact sequence of MCM R-modules $0 \to L \to M \to N \to 0$ and an MCM R-module W, from 3.4.5, one has a long exact sequence of R-modules

$$\cdots \longrightarrow \widehat{\mathrm{Ext}}^{i-1}_R(W,L) \longrightarrow \widehat{\mathrm{Ext}}^{i}_R(W,L) \longrightarrow \widehat{\mathrm{Ext}}^{i}_R(W,M) \longrightarrow \widehat{\mathrm{Ext}}^{i}_R(W,N) \longrightarrow \widehat{\mathrm{Ext}}^{i+1}_R(W,L) \longrightarrow \cdots$$

Example 3.4.9. Note that if R is regular, then $\underline{\mathsf{MCM}}(R)$ is zero. The converse holds when R is Cohen-Macaulay.

Example 3.4.10. Let $R = k[x]/(x^2)$ with k a field. In this setting, the unique indecomposable non-free MCM R-module is k and here there is an infinite exact sequence

$$\cdots \xrightarrow{x\cdot} R \xrightarrow{x\cdot} R \xrightarrow{x\cdot} R \xrightarrow{\cdot} \cdots\,.$$

In particular, $k \cong \Omega^n k$ for any $k \in \mathbb{Z}$. It then follows that $\widehat{\mathrm{Ext}}^n_R(k,k) = k$ for all $n \in \mathbb{Z}$.

3.5 Acyclic complexes in exact categories

Let B be a Frobenius category and $\mathsf{K}(\mathsf{B})$ the homotopy category associated to B; that is, $\mathsf{K}(\mathsf{B})$ is the stabilization of the Frobenius exact category of complexes $\mathsf{C}(\mathsf{B})$ over B, where the phantom maps are exactly the null-homotopic maps. [2]

Definition 3.5.1. A complex A over B is *acyclic* if there exist exact sequences

$$0 \longrightarrow Z_{i+1} \xrightarrow{a_i} A_i \xrightarrow{b_i} Z_i \longrightarrow 0 \quad \text{for } i \in \mathbb{Z}$$

in B such that $\partial_i^A = a_{i-1}b_i$.

Remark 3.5.2. Let A be acyclic. It is not hard to verify that such a family of exact sequences witnessing the acyclicity of A, as in Definition 3.5.1, is well-defined up to isomorphism. Furthermore, and if $A \to A'$ is a morphism of acyclic complexes, there is a well-defined induced map on the corresponding exact sequences. With this in mind, we fix, for each acyclic complex A, a family of exact sequence as above, and write $Z_i(A)$ for the corresponding Z_i.

3.5.3. Let $\mathsf{C}_{\mathrm{ac}}(\mathsf{B})$ denote the full subcategory of $\mathsf{C}(\mathsf{B})$ consisting of acyclic complexes. It is an additive subcategory, so can be viewed as exact category, with the induced exact structure. The exact sequences in $\mathsf{C}_{\mathrm{ac}}(\mathsf{B})$ are degreewise split-exact sequences of acyclic complexes.

Lemma 3.5.4. *For each $i \in \mathbb{Z}$, the assignment $A \mapsto Z_i(A)$ is exact functor from $\mathsf{C}_{\mathrm{ac}}(\mathsf{B}) \to \mathsf{B}$. That is to say, it is an additive functor that preserves admissible exact sequences.*

Proof. This is a routine verification, using the Snake Lemma for exact categories. □

3.5.5. Recall that $\mathsf{proj}(\mathsf{B})$ denotes the projective objects in B. It is an additive subcategory of B, and hence the corresponding category of complexes, and its homotopy category $\mathsf{K}(\mathsf{proj}(\mathsf{B}))$ can be regarded as full subcategories of $\mathsf{C}(\mathsf{B})$ and $\mathsf{K}(\mathsf{B})$, respectively. In the same vein, $\mathsf{C}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B}))$, the acyclic complexes over $\mathsf{proj}(\mathsf{B})$, and $\mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B}))$, the associated homotopy category, are full subcategories of $\mathsf{C}_{\mathrm{ac}}(\mathsf{B})$ and $\mathsf{K}_{\mathrm{ac}}(\mathsf{B})$, respectively. Thus one has inclusions of triangulated categories

$$\mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B})) \subseteq \mathsf{K}_{\mathrm{ac}}(\mathsf{B}) \subseteq \mathsf{K}(\mathsf{B})\,.$$

Theorem 3.5.6. *For each $i \in \mathbb{Z}$, the restriction of the exact functor $\mathsf{C}_{\mathrm{ac}}(\mathsf{B}) \to \mathsf{B}$ to $\mathsf{C}_{\mathrm{ac}}(\mathsf{proj}(B))$, given by $A \mapsto Z_iA$, induces an equivalence of triangulated categories $Z_i \colon \mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(B)) \to \underline{\mathsf{B}}$.*

Proof. Without loss of generality, we can assume $i = 0$, and we write $Z = Z_0$. Furthermore, the restriction of Z on $\mathsf{C}_{\mathrm{ac}}(\mathsf{B})$ to $\mathsf{C}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B}))$ is still an exact functor between Frobenius exact categories. Therefore, by 3.4.4, it suffices to show $Z(\mathsf{proj}(\mathsf{C}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B})))) \subseteq \mathsf{proj}(\mathsf{B})$ and $Z(\Sigma -) \simeq \Omega^{-1}Z(-)$ to get a well-defined induced functor on triangulated categories $Z \colon \mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(B)) \to \underline{\mathsf{B}}$.

To this end, let $A \in \mathsf{C}_{\mathrm{ac}}(\mathsf{proj}(\mathsf{B}))$ and consider $C = \operatorname{cone}(\mathrm{id}_A)$. Recall that

$$A \cong \bigoplus_{n\in\mathbb{Z}} \operatorname{cone}(\mathrm{id}_{\Sigma^n A_n}) = \cdots A_1 \oplus A_0 \xrightarrow{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}} A_0 \oplus A_{-1} \xrightarrow{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}} A_{-1} \oplus A_{-2} \to \cdots$$

and hence, $ZC = A_0 \in \mathsf{proj}(\mathsf{B})$. This verifies the necessary containment. Also, the exact sequence

$$0 \to ZA \to A_{-1} \to Z_{-1}A \to 0$$

coming from the fact A is acyclic, also gives a natural isomorphism $Z_{-1}A \cong \Omega^{-1}ZA$ in $\underline{\mathsf{B}}$ by Construction 3.3.6. Finally, note $Z(\Sigma A) = Z_{-1}A$. Therefore applying 3.4.4 yields an exact functor of triangulated categories $Z \colon \mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(B)) \to \underline{\mathsf{B}}$.

[2] The fact $\mathsf{C}(\mathsf{B})$ is Frobenius exact was Problem 2 on Discussion Problems 3.

First, we show it is essentially surjective. Indeed, if X is in $\underline{\mathsf{B}}$ then one can inductive (in both directions) construct an infinite acyclic complex of projectives A where

$$0 \to \Omega^{i+1}X \to A_i \to \Omega^i X \to 0$$

are the defining exact sequences making A acyclic. Then by construction $ZA = X$.

To see Z is fully faithful, observe for A, A' acyclic complex of projectives we factor the desired isomorphism (the unlabeled map below) as

$$\begin{array}{ccc} \mathrm{Hom}_{\mathsf{K}}(A, A') & \longrightarrow & \underline{\mathrm{Hom}}_{\mathsf{B}}(Z(A), Z(A')) \\ {\scriptstyle\alpha}\downarrow & & \uparrow{\scriptstyle\gamma} \\ \mathrm{Hom}_{\mathsf{K}}(A, A'_{\geqslant 0}) & \xrightarrow[\beta]{} & \mathrm{Hom}_{\mathsf{K}}(A, Z(A')) \end{array} \tag{3.5.6.1}$$

where α is induced by the canonical truncation $A' \to A'_{\geqslant 0}$, the map β by the augmentation

$$A'_{\geqslant 0} \twoheadrightarrow Z_{-1}(A') = Z(A')\,,$$

and γ by the functor Z. We verify α, β, γ are isomorphisms.

Note that for any $C, C' \in \underline{\mathsf{C}}(\mathsf{B})$ we can define a complex of abelian groups:

$$\mathcal{H}om(C, C')_n := \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathsf{B}}(C_i, C_{i+n}) \quad \text{and differential} \quad (h_i)_{i\in\mathbb{Z}} \mapsto (\partial' h_i - (-1)^n h_{i-1}\partial)_{i\in\mathbb{Z}}\,.$$

It is straightforward to verify that this is indeed a complex and that

$$\mathrm{Hom}_{\mathsf{K}}(C, \Sigma^n C') \cong \mathrm{H}^n(\mathcal{H}om(C, C')) \quad \text{for } n \in \mathbb{Z}.$$

In our context, to show α is an isomorphism we prove the following:

Claim. For any complex C in $\mathsf{C}(\mathsf{proj}(\mathsf{B}))$ with $C_i = 0$ for $i \gg 0$, one has $\mathrm{Hom}^*_{\mathsf{K}}(A, C) = 0$.

To verify the claim, we first show this when C is bounded. Here we can induct on the length of C and so it suffices to verify the claim when $C \in \mathsf{proj}(\mathsf{B})$. The point is that $\mathcal{H}om(A, C)$ has the form

$$\cdots \to \mathrm{Hom}_{\mathsf{B}}(A_{-1}, C) \to \mathrm{Hom}_{\mathsf{B}}(A_0, C) \to \mathrm{Hom}_{\mathsf{B}}(A_1, C) \to \cdots$$

which can be spliced together by the short exact sequences

$$0 \to \mathrm{Hom}_{\mathsf{B}}(Z_i A, C) \to \mathrm{Hom}_{\mathsf{B}}(A_i, C) \to \mathrm{Hom}_{\mathsf{B}}(Z_{i+1}A, C) \to 0$$

for each $i \in \mathbb{Z}$; this is usually only left exact, but because C is projective these are short exact. In particular, $\mathcal{H}om(A, C)$ is acyclic and so we have verified the claim when C is bounded. More generally, C is the limit of the following

$$\cdots \xrightarrow{\sigma_2} C_{\geqslant 1} \xrightarrow{\sigma_1} C_{\geqslant 0} \xrightarrow{\sigma_0} C_{\geqslant -1} \xrightarrow{\sigma_{-1}} \cdots$$

and in particular, there is a short exact sequence of complex

$$0 \to C \to \prod_{n\in\mathbb{Z}} C_{\geqslant n} \to \prod_{n\in\mathbb{Z}} C_{\geqslant n} \to 0$$

where surjective map is given by $(c_n) \mapsto (c_n - \sigma_{n+1}(c_{n+1}))$. This defines the exact sequence of chain complexes of abelian groups

$$0 \to \mathcal{H}om(A, C) \to \prod_{n\in\mathbb{Z}} \mathcal{H}om(A, C_{\geqslant n}) \to \prod_{n\in\mathbb{Z}} \mathcal{H}om(A, C_{\geqslant n}) \to 0\,.$$

Now as each $C_{\geqslant n}$ is a bounded complex of projectives, it follows from the previously established case, that each $\mathcal{H}om(A, C_{\geqslant n})$ is acyclic. Therefore, the short exact sequence directly above forces $\mathcal{H}om(A, C)$ to be acyclic, finishing the proof of the claim.

Now applying the claim to the $A'_{<0}$ we have that $\mathrm{Hom}^*_{\mathsf{K}}(A, A'_{<0}) = 0$. Furthermore, α fits into the exact sequence

$$0 = \mathrm{Hom}_{\mathsf{K}}(A, A'_{<0}) \to \mathrm{Hom}_{\mathsf{K}}(A, A') \xrightarrow{\alpha} \mathrm{Hom}_{\mathsf{K}}(A, A'_{\geqslant 0}) \to \mathrm{Hom}_{\mathsf{K}}(A, \Sigma A'_{<0}) = 0\,,$$

where the sequence arises from applying $\mathrm{Hom}_{\mathsf{K}}(A, -)$ to the exact triangle $A'_{<0} \to A' \to A'_{\geqslant 0}$, and the outer terms are zero by the already remarked fact. Thus, α is an isomorphism.

For β, observe that there is an exact sequence of complexes

$$0 \to C \to A'_{\geqslant 0} \to ZA \to 0$$

and note that since $A'_{\geqslant}$ is a projective resolution of ZA, then C is an acyclic nonnegatively graded complex. The latter forces,

$$\mathrm{Hom}_{\mathsf{K}}(A, C) \cong \mathrm{Hom}_{\mathsf{K}}(A_{\geqslant -1}, C)$$

and now since C acyclic and $A_{\geqslant -1}$ is a complex of projectives, it follows that the right-hand side is zero. Similarly, it follows that $\mathrm{Hom}_{\mathsf{K}}(A, \Sigma C) = 0$. Therefore, defining exact sequence of C determine the exact sequence of abelian groups

$$0 = \mathrm{Hom}_{\mathsf{K}}(A, C) \to \mathrm{Hom}_{\mathsf{K}}(A, A_{\geqslant -1}) \xrightarrow{\beta} \mathrm{Hom}_{\mathsf{K}}(A, ZA') \to \mathrm{Hom}_{\mathsf{K}}(A, \Sigma C) = 0\,,$$

finishing the proof that β is an isomorphism.

Finally, for the analysis of γ, we have that γ is surjective. For converse, let $f\colon A \to ZA'$ be a map of complexes such that the induced map $\gamma(f)\colon ZA \to ZA'$ is zero in $\underline{\mathsf{B}}$. Then $\gamma(f)$ factors through a projective but using that $ZA \to A_{-1}$ is an admissible mono, it follows that $\gamma(f)$ factors through ∂_0^A. However, this exactly says that the map f is null-homotopic. Hence, γ is injective.

Therefore, α, β, γ are isomorphisms as claimed and hence the map on top in (3.5.6.1) is an isomorphism. Hence, Z is fully faithful, which finishes the proof it induces an equivalence on stable categories. □

Remark 3.5.7. The quasi-inverse of Z was described when proving Z was essentially surjective. Namely, let $\mathsf{cp}\colon \underline{\mathsf{B}} \to \mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(\underline{\mathsf{B}}))$ is given by the assignment that associative to M the acyclic complex of projectives

$$\mathsf{cp}M := \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P_{-1} \longrightarrow P_{-2} \longrightarrow \cdots$$

with $\Omega(M)$, M, $\Omega^{-1}M$ the corresponding syzygies.

and using the lifting property to the left (and to the right) one can functorially lift maps in $\underline{\mathsf{B}}$ uniquely up to homotopy on such complexes. We call $\mathsf{cp}M$ a **complete (projective) resolution** of M.

Specializing Theorem 3.5.6 to $\mathsf{B} := \mathsf{MCM}(R)$, the category of MCM modules over a Gorenstein ring yields the following result.

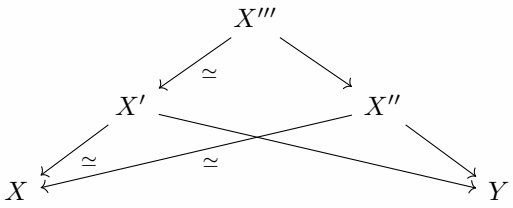
Corollary 3.5.8. *Let R be a commutative Gorenstein ring. The functor $Z\colon \mathsf{K}_{\mathrm{ac}}(\mathsf{proj}(R)) \to \underline{\mathsf{MCM}}(R)$ is an equivalence of triangulated categories. Furthermore, for MCM R-modules M, N,*

$$\widehat{\mathrm{Ext}}^i_R(M, N) \cong \mathrm{Hom}_{\mathsf{K}(R)}(\mathsf{cp}(M), \Sigma^i N) = \mathrm{H}^i(\mathrm{Hom}_R(\mathsf{cp}(M), N)) \quad \textit{for all } i \in \mathbb{Z}\,.$$ □

3.6 The singularity category

Definition 3.6.1. Let S be a thick subcategory of a triangulated category T. We say a morphism $f\colon X \to Y$ is an **S-isomorphism** provided that $\mathrm{cone}(f)$ is an object of S. We denote an S-isomorphism by $\simeq$.

The **Verdier localization** of T with respect to S is the category T/S whose objects are exactly the objects of T and morphisms are diagrams $X \xleftarrow{\simeq} X' \to Y$ with the equivalence relation given $X \xleftarrow{\simeq} X' \to Y$ is equivalent to $X \xleftarrow{\simeq} X'' \to Y$ if these fit into the following commutative diagram:



Confer Construction 1.3.5. The images of triangles from T are triangles in T/S, making the latter a triangulated category (with the induced suspension functor).

Let R be a commutative noetherian ring.

Definition 3.6.2. The **singularity category of** R is $\mathsf{D}_{\mathsf{sg}}(R) = \mathsf{D}^{\mathsf{b}}(\operatorname{\mathsf{mod}} R)/\operatorname{\mathsf{Perf}} R$

Lemma 3.6.3. *For a morphism $f\colon X \to Y$ in $\mathsf{D}(R)$, its image in $\mathsf{D}_{\mathsf{sg}}(R)$ is zero if and only if f factors through a perfect complex in $\mathsf{D}(R)$. In particular, X is zero in $\mathsf{D}_{\mathsf{sg}}(R)$ if and only if X is in $\operatorname{\mathsf{Perf}} R$.*

Proof. The "in particular" statement is immediate from the first statement, so we prove the latter. Note that a morphism $g\colon X \to Y$ in $\mathsf{D}(R)$ has as its image in $\mathsf{D}_{\mathsf{sg}}(R)$ the fraction

$$X \xleftarrow{=} X \xrightarrow{g} Y\,.$$

Hence, for the forward direction, there exists a morphism $g\colon X' \to X$ in $\mathsf{D}(R)$ such that $fg = 0$ and $\operatorname{cone}(g)$ is a perfect complex. Now it is straightforward to see that f factors through $\operatorname{cone}(g)$.

Conversely, assume f factors through a perfect complex P. That is, the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{g} & P \\ \downarrow{\scriptstyle f} & & \downarrow \\ Y & \xrightarrow[=]{} & Y \end{array}$$

and hence there is a morphism of triangles

$$\begin{array}{ccccccc} X' & \xrightarrow{g} & X & \longrightarrow & P & \longrightarrow & \Sigma X' \\ \downarrow & & \downarrow{\scriptstyle f} & & \downarrow & & \downarrow \\ 0 & \longrightarrow & Y & \xrightarrow[=]{} & Y & \longrightarrow & 0 \end{array}$$

Therefore, it follows that f is zero in $\mathsf{D}_{\mathsf{sg}}(R)$. □

Example 3.6.4. R is a regular ring if and only if $\mathsf{D}_{\mathsf{sg}}(R) = 0$. Indeed, R is regular if and only if $\mathsf{D}_{\mathsf{sg}}(R_{\mathfrak{p}}) = 0$ for all $\mathfrak{p} \in \operatorname{Spec} R$. As $\mathsf{D}_{\mathsf{sg}}(R) \to \mathsf{D}_{\mathsf{sg}}(R_{\mathfrak{p}})$ is essentially surjective, it remains to verify if $M_{\mathfrak{p}}$ is in $\operatorname{\mathsf{Perf}} R_{\mathfrak{p}}$ for all $\mathfrak{p} \in \operatorname{Spec} R$, then M is in $\operatorname{\mathsf{Perf}} R$. Letting $F \xrightarrow{\simeq} M$ be a semifree resolution with F_i finite rank free, we set $C_n = \operatorname{coker}\partial^F_n$ and define

$$U_n = \{\mathfrak{p} \in \operatorname{Spec} R : (C_n)_{\mathfrak{p}} \text{ is projective}\}\,.$$

It follows that $U_n \subseteq U_{n+1}$ and that each U_n is an open subset of $\operatorname{Spec} R$. Furthermore $\cup U_n = \operatorname{Spec} R$, by assumption, and so the quasi-compactness of $\operatorname{Spec} R$ implies that $U_n = \operatorname{Spec} R$ for all $n \gg 0$. In particular C_n is a projective R-module for all $n \gg 0$, and so M is in $\operatorname{\mathsf{Perf}} R$.

Lemma 3.6.5. *Assume M is a finitely generated R-module such that $\operatorname{Ext}^{>0}_R(M, R) = 0$. For each perfect complex with $P_i = 0$ for all $i \leq 0$, one has*

$$\operatorname{Hom}_{\mathsf{D}(R)}(M, P) = 0.$$

Proof. Induct on the length of P. When $P \simeq \Sigma^i P_i$ where P_i is in $\mathsf{proj}(R)$ for some $i > 0$, then

$$\operatorname{Hom}_{\mathsf{D}(R)}(M, P) \cong \operatorname{Hom}_{\mathsf{D}(R)}(M, \Sigma^i P_i) = \operatorname{Ext}^i_R(M, P_i)$$

where the last module is zero by assumption. Now assume $P = P_{\geqslant i}$ where $i > 0$ and $P_i \neq 0$. Consider the exact triangle

$$\Sigma^i P_i \to P \to P_{>i} \to \Sigma^{i+1} P_i$$

and so applying $\operatorname{Hom}_{\mathsf{D}(R)}(M, -)$ yields an exact sequence of R-modules

$$\operatorname{Ext}^i_R(M, P_i) \to \operatorname{Hom}_{\mathsf{D}(R)}(M, P) \to \operatorname{Hom}_{\mathsf{D}(R)}(M, P_{>i}) \to \operatorname{Ext}^{i+1}_R(M, P_i)$$

and so by induction it follows that $\operatorname{Hom}_{\mathsf{D}(R)}(M, P) = 0$, as claimed. □

Lemma 3.6.6. *Assume M is a finitely generated R-module such that $\operatorname{Ext}^{>0}_R(M, R) = 0$. If $f\colon M \to N$ is morphism in $\mathsf{mod}(R)$ whose image is zero in $\mathsf{D}_{\mathsf{sg}}(R)$, then f is a phantom map in $\mathsf{mod}(R)$ (i.e., f factors through an honest finitely generated projective R-module in $\mathsf{mod}(R)$).*

Proof. By Lemma 3.6.3, f factors as $M \xrightarrow{\alpha} P \xrightarrow{\beta} N$ in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ where P is in $\mathsf{Perf}\, RR$. As P is perfect, β can be chosen to be a morphism of complexes. For degree reasons, β induces a morphism of complexes $P_{\geqslant 0} \xrightarrow{\beta} N$. Also, by degree reasons α induces a morphisms $M \xrightarrow{\alpha} P \to P_{\geqslant 0}$. Furthermore, by Lemma 3.6.5

$$\begin{array}{ccc} M & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ P_{\geqslant 0} & \longrightarrow & P_{\geqslant 1} \end{array}$$

commutes in $\mathsf{D}(R)$ and so we get a morphism of triangles

$$\begin{array}{ccccccc} M & \xrightarrow{=} & M & \longrightarrow & 0 & \longrightarrow & \Sigma M \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ P_0 & \longrightarrow & P_{\geqslant 0} & \longrightarrow & P_{\geqslant 1} & \longrightarrow & \Sigma P_0 \end{array}$$

Therefore,

$$M \to P_0 \to P_{\geqslant 0} \xrightarrow{\beta} N$$

is a factorization of f and so $M \to P_0 \to N$ is a factorization of f in $\mathsf{mod}(R)$. □

Lemma 3.6.7. *Let M be a finitely generated R-module with $\operatorname{Ext}^{>0}_R(M, R) = 0$. If N is a finitely generated R-module, then the natural map $\operatorname{Hom}_R(M, N) \to \operatorname{Hom}_{\mathsf{D}_{\mathsf{sg}}(R)}(M, N)$ is surjective.*

Proof. Fix a morphism $fu^{-1}\colon M \to N$ in $\mathsf{D}_{\mathsf{sg}}(R)$. This can be represented by the diagram $M \xleftarrow{u} X \xrightarrow{f} N$ in $\mathsf{D}(R)$ with $P := \operatorname{cone}(u)$ perfect; we may assume P is a bounded complex of finitely generated projective modules. The exact sequence

$$0 \longrightarrow P_{\leqslant 0} \longrightarrow P \longrightarrow P_{\geqslant 1} \longrightarrow 0$$

induces the triangle in the top row of the following commutative diagram in $\mathsf{D}(R)$:

$$\begin{array}{ccccccc} \Sigma^{-1}(P_{\leqslant 0}) & \longrightarrow & \Sigma^{-1}P & \longrightarrow & \Sigma^{-1}(P_{\geqslant 1}) & \longrightarrow & \\ \| & & \downarrow & & \vdots & & \\ \Sigma^{-1}(P_{\leqslant 0}) & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & \\ & & {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} & \searrow{\scriptstyle g} & \\ & & M & = & M & & N \\ & & \downarrow & & \downarrow & & \\ & & P & \longrightarrow & P_{\geqslant 1} & & \end{array}$$

The second column from the left is the triangle associated to u, the top left square commutes by construction, and the second row is a triangle induced by the morphism $\Sigma^{-1}(P_{\leqslant 0}) \to X$. The third column is a triangle induced by the octahedral axiom. The map g is induced by the map $f\colon X \to N$, since

$$\operatorname{Hom}_{\mathsf{D}}(\Sigma^{-1}(P_{\leqslant 0}), N) \cong \operatorname{Hom}_{\mathsf{K}}(\Sigma^{-1}(P_{\leqslant 0}), N) = 0$$

where the equality holds for degree reasons.

The diagram above shows that the cone of the map $X \to Y$ is perfect, and hence also that $fu^{-1} = gv^{-1}$ in $\mathsf{D}_{\mathsf{sg}}(R)$. Since $\operatorname{Hom}_{\mathsf{D}}(M, P_{\geqslant 1}) = 0$, by Lemma 3.6.5, and $\operatorname{cone}(v) = P_{\geqslant 0}$, by construction, the map $v\colon Y \to M$ has a section, say s. It is easy to verify now the $gv^{-1} = gs$ in $\mathsf{D}_{\mathsf{sg}}(R)$. It remains to note that $gs\colon M \to N$ is equivalent to the R-linear map $\mathrm{H}_0(gs)\colon M \to N$, which completes the proof. □

In the remainder of the section the ring R will be Gorenstein. Phantom maps in $\mathsf{mod}(R)$ become 0 in $\mathsf{D}_{\mathsf{sg}}(R)$, by Lemma 3.6.3, so the composition $\mathsf{mod}(R) \to \mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R) \to \mathsf{D}_{\mathsf{sg}}(R)$ induces an additive functor $\underline{\mathsf{mod}}(R) \to \mathsf{D}_{\mathsf{sg}}(R)$. Restricting this to the subcategory of MCM modules yields a functor

$$F\colon \underline{\mathsf{MCM}}(R) \longrightarrow \mathsf{D}_{\mathsf{sg}}(R)\,.$$

Observe that the source and target of this functor are triangulated categories.

Theorem 3.6.8. *When R is Gorenstein, the functor F is an equivalence of triangulated categories.*

Proof. For F to be triangulated it suffices to check that $F(\Omega^{-1}-) = \Sigma F(-)$, as F is already additive and the original functor $\mathsf{MCM}(R) \to \mathsf{D}_{\mathsf{sg}}(R)$ sent exact sequences to exact triangles.

To this end, let C be a complete resolution of M and consider $P' = C_{\geqslant -1}$ and $P_{\geqslant 0}$. Then $P \xrightarrow{\simeq} M$ and $P' \xrightarrow{\simeq} \Sigma^{-1}\Omega_R^{-1}(M)$. The triangle $P_{-1} \to P' \to P \to \Sigma P_{-1}$ in $\mathsf{D}^{\mathsf{b}}(\mathsf{mod}\, R)$ defines an isomorphism

$$\Sigma^{-1}\Omega_R^{-1}(M) \simeq P' \simeq P \simeq M$$

but this is exactly showing that $F(\Omega^{-1}M) \simeq \Sigma F(M)$.

Next, by Lemmas 3.6.6 and 3.6.7 it follows that F is a fully faithful. Hence, it remains to check F is essentially surjective. Let X be an object in $\mathsf{D}_{\mathsf{sg}}(R)$; we can assume that it is a complex of finitely generated projective R-modules with $X_i = 0$ for $i \ll 0$. Then for any integer n the complex $X_{<n}$ is perfect, and hence it follows from the exact sequence of complexes

$$0 \longrightarrow X_{<n} \longrightarrow X \longrightarrow X_{\geqslant n} \longrightarrow 0$$

that the natural map $X \to X_{\geqslant n}$ is an isomorphism in $\mathsf{D}_{\mathsf{sg}}(R)$. For $s = \sup X - 1$ the complex $X_{\geqslant s}$ has homology only in degree s, so that

$$X \cong X_{\geqslant s} \cong \Sigma^s \mathrm{H}_s(X_{\geqslant s})\,, \qquad \text{in } \mathsf{D}_{\mathsf{sg}}(R).$$

It thus suffices to verify that any R-module, say L, is isomorphic, in $\mathsf{D}_{\mathsf{sg}}(R)$, to an MCM module. Since R is Gorenstein, some syzygy module, $\Omega^d(L)$, is MCM, by Theorem 2.2.8. It remains to observe that if P is a projective resolution of L then there are isomorphisms

$$L \cong P \cong P_{\geqslant d} \cong \Sigma^d \Omega^d(L)$$

where the first and last one hold already in $\mathsf{D}(R)$ and the middle one holds in $\mathsf{D}_{\mathsf{sg}}(R)$, as explained above. □

Chapter 4

Serre duality on the stable category

4.1 Serre duality

Fix a commutative noetherian ring R, and let E denote an injective R-module. Also, set $D = \operatorname{Hom}_R(-, E)$ viewed as an endofunctor on $\mathsf{Mod}(R)$. Let A be an R-linear additive category. That is to say, A is an additive category where for each A in A there is a homomorphism of rings

$$\varphi_A \colon R \longrightarrow \operatorname{Hom}_{\mathsf{A}}(A, A)$$

such that for B in A the induced left and right actions on $\operatorname{Hom}_{\mathsf{A}}(A, B)$ via φ_B and φ_A, respectively, coincide. We will be particularly interested in the case that A is $\mathsf{D}_{\mathsf{sg}}(R)$ when R is Gorenstein.

A **Serre functor** on A (with respect to D) is an R-linear equivalence $S \colon \mathsf{A} \to \mathsf{A}$ with isomorphisms

$$\eta_{A,B} \colon D \operatorname{Hom}_{\mathsf{A}}(A, B) \xrightarrow{\cong} \operatorname{Hom}_{\mathsf{A}}(B, SA) \tag{4.1.0.1}$$

for A, B in A that are functorial in both arguments. The phrases "A has a Serre functor" and "A has Serre duality" will be used interchangeably.

Remark 4.1.1. The existence of a Serre functor is equivalent to the representability of the contravariant functor

$$D \operatorname{Hom}_{\mathsf{A}}(A, -) \colon \mathsf{A} \longrightarrow \mathsf{Mod}(R)\,,$$

for each A in A. This perspective makes it clear that if A admits a Serre functor, then the functor is unique up to unique isomorphism.

Remark 4.1.2. Let $S \colon \mathsf{A} \to \mathsf{A}$ be an R-linear equivalence of categories. When S is a Serre functor, corresponding to the identity map in $\operatorname{Hom}_{\mathsf{A}}(SA, SA)$, there is an R-linear *trace map*

$$\operatorname{tr}_A \colon \operatorname{Hom}_{\mathsf{A}}(A, SA) \longrightarrow E$$

and this defines a non-degenerate R-bilinear pairing

$$\operatorname{Hom}_{\mathsf{A}}(A, B) \times \operatorname{Hom}_{\mathsf{A}}(B, SA) \longrightarrow \operatorname{Hom}_{\mathsf{A}}(A, SA) \xrightarrow{\operatorname{tr}_A} E$$

where the first map is the usual composition. The naturality of the Serre functor means that for any map $\alpha \colon A \to A'$, one has

$$\operatorname{tr}_A(g \circ (f \circ \alpha)) = \operatorname{tr}_{A'}((S\alpha \circ g) \circ f)$$

for any $f \in \operatorname{Hom}_{\mathsf{A}}(A', B)$ and $g \in \operatorname{Hom}_{\mathsf{A}}(B, SA)$.

Conversely, given a coherent family of trace maps $\{\operatorname{tr}_A\}_{A \in \mathsf{A}}$ such that the pairing defined as above is non-degenerate, the functor S is a Serre functor on A. The coherence condition is that

Thus, the data of a Serre functor is equivalent to that of a coherent family of trace maps on A.

Example 4.1.3. If R is a field and $E = R$, then $D = \operatorname{Hom}_R(-, R) = (-)^*$. In this case, $\mathsf{mod}(R)$ has Serre duality with $S = \mathrm{id}$. Indeed,

$$D \operatorname{Hom}_R(M, N) \cong D \operatorname{Hom}_R(M \otimes_R N^*, R) \cong M \otimes_R N^* \cong \operatorname{Hom}_R(N, M) .$$

Example 4.1.4. Let k be a field and X a smooth projective scheme over k, of dimension d. Let ω_X be the canonical sheaf on X. Classical Serre duality can be interpreted as the statement that $\Sigma^d \omega_X \otimes (-)$ is the Serre functor on D, the bounded derived category of coherent sheaves on X, viewed as a k-linear category. That is to say, for $\mathcal{F}, \mathcal{G}$ in D there are natural isomorphisms

$$\operatorname{Hom}_k(\operatorname{Hom}_{\mathsf{D}}(\mathcal{F}, \mathcal{G}), k) \xrightarrow{\cong} \operatorname{Hom}_{\mathsf{D}}(\mathcal{G}, \Sigma^d \omega_X \otimes \mathcal{F}) .$$

4.2 Auslander–Reiten formula

As before let R be a commutative noetherian ring and E an injective R-module. We set $(-)^* = \operatorname{Hom}_R(-, R)$ and $D = \operatorname{Hom}_R(-, E)$ viewed as endofunctors on $\mathsf{Mod}(R)$.

Definition 4.2.1. Let M be in $\mathsf{mod}(R)$, and consider a projective presentation

$$P_1 \xrightarrow{f} P_0 \longrightarrow X \longrightarrow 0$$

where each P_i is a finitely generated R-module. The *Auslander transpose*, or just the *transpose*, of M is the R-module determined by the diagram induced by applying $(-)^*$ to the presentation above:

$$P_0^* \xrightarrow{f^*} P_1^* \longrightarrow \operatorname{Tr}_R(M) \longrightarrow 0 .$$

Remark 4.2.2. When the P_i are finite free modules, then f can be represented by a matrix and its transpose represents f^*, hence the nomenclature. Also, the transpose is well-defined up to projective summands.

4.2.3. Fix an exact sequence of R-modules:

$$\xi \colon 0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0 .$$

The induces functors $\xi^*, \xi_* \colon \mathsf{Mod}(R) \to \mathsf{Mod}(R)$ defined by exact sequences

$$0 \longrightarrow \operatorname{Hom}_R(-, A) \longrightarrow \operatorname{Hom}_R(-, B) \longrightarrow \operatorname{Hom}_R(-, C) \longrightarrow \xi^* \longrightarrow 0$$
$$0 \longrightarrow \operatorname{Hom}_R(C, -) \longrightarrow \operatorname{Hom}_R(B, -) \longrightarrow \operatorname{Hom}_R(A, -) \longrightarrow \xi_* \longrightarrow 0$$

As the notation suggests, ξ^* is contravariant whereas ξ_* is covariant.

The following theorem[1] of Auslander is called the *defect formula*.

Theorem 4.2.4. *For a finitely generated R-module M, there is an isomorphism $D\xi^*(M) \cong \xi_*(D \operatorname{Tr}_R(M))$, natural in M and ξ.*

The main result of the section is the following theorem of Auslander–Reiten, called the **AR-formula**.

Theorem 4.2.5. *For any finitely generated R-module M and arbitrary R-module C, there is a natural isomorphism of R-modules*

$$D\underline{\operatorname{Hom}}_R(M, C) \cong \operatorname{Ext}^1_R(C, D \operatorname{Tr}_R(M)) .$$

[1] This proof is left to Exercise 3 on Discussion Problems 4

Proof. Consider an exact sequence $\xi\colon 0 \to \Omega C \to P \to C \to 0$ of R-modules, where P is a projective. It follows from the exact sequence of R-modules

$$0 \longrightarrow \operatorname{Hom}_R(M, \Omega C) \longrightarrow \operatorname{Hom}_R(M, P) \longrightarrow \operatorname{Hom}_R(M, C) \longrightarrow \xi^*(M) \longrightarrow 0$$

that $\xi^*(M) \cong \underline{\operatorname{Hom}}_R(M, C)$. On the other hand, applying $\operatorname{Hom}_R(-, D\operatorname{Tr}_R(M))$ to ξ one gets that

$$\xi_*(D\operatorname{Tr}_R(M)) = \operatorname{Ext}^1_R(C, D\operatorname{Tr}_R(M))\,.$$

The AR formula is thus a direct consequence of the defect formula, Theorem 4.2.4. □

Example 4.2.6. Let R be artinian Gorenstein. Then $\mathsf{MCM}(R) = \mathsf{mod}(R)$ and Theorem 4.2.5 specializes (with $R = E$) to an isomorphism:

$$\operatorname{Hom}_R(\underline{\operatorname{Hom}}_R(M, N), R) \cong \underline{\operatorname{Hom}}_R(N, \Omega^{-1} D\operatorname{Tr}_R(M)) \quad \text{for all } M, N \text{ in } \mathsf{mod}(R).$$

Said otherwise, $\Omega^{-1}D\operatorname{Tr}_R(-)$ is a Serre functor on $\underline{\mathsf{mod}}(R)$. Furthermore, $D\operatorname{Tr}_R(-) = \Omega^2$ and so Ω is a Serre functor on $\underline{\mathsf{MCM}}(R)$. That is, there are natural isomorphisms

$$\operatorname{Hom}_R(\underline{\operatorname{Hom}}_R(M, N), R) \cong \underline{\operatorname{Hom}}_R(N, \Omega_R M)\,.$$

4.3 Serre duality on $\underline{\mathsf{MCM}}(R)$ via Auslander duality

From now on, $(R, \mathfrak{m}, k)$ is a Gorenstein local ring that is an isolated singularity. The latter condition means that the ring R is regular on the punctured spectrum: The local ring $R_\mathfrak{p}$ is regular for each $\mathfrak{p}$ in $\operatorname{Spec} R \setminus \{\mathfrak{m}\}$. The main result is the following due to Auslander which extends Example 4.2.6.

Theorem 4.3.1. *When $(R, \mathfrak{m}, k)$ is an isolated Gorenstein singularity, there are natural isomorphisms*

$$D\underline{\operatorname{Hom}}_R(M, N) \cong \underline{\operatorname{Hom}}_R(N, \Omega^{1-d}(M)) \quad \textit{for all } M, N \in \mathsf{MCM}(R),$$

where d is the Krull dimension and $D := \operatorname{Hom}_R(-, E_R(k))$. In particular, $\underline{\mathsf{MCM}}(R)$ has Serre duality with functor $\Omega^{1-d}(-)$.

Before beginning the proof, we note the following ingredients.

4.3.2. As R is Gorenstein, we have that $\mathrm{H}^i_\mathfrak{m}(R) = \begin{cases} E_R(k) & i = \dim R \\ 0 & \text{else} \end{cases}$; indeed, this follows as $R = \omega_R$ and the discussion on dualizing complexes in Section 1.6. Also, since R has an isolated singularity, it is straightforward to see that $\underline{\operatorname{Hom}}_R(M, N)$ is a finite length R-module for any pair of MCM R-modules M, N.[2]

Also, we need the following general result on local cohomology.

Lemma 4.3.3. *Let $(R, \mathfrak{m}, k)$ be a local ring of dimension d and assume that M is an arbitrary R-complex satisfying that $\mathrm{H}^i(M)$ is a finite length R-module for each $i > 0$. Then the map $\mathrm{R}\Gamma_\mathfrak{m}(M) \to M$ induces isomorphisms $\mathrm{H}^i_\mathfrak{m}(M) \xrightarrow{\cong} \mathrm{H}^i(M)$ for $i > d$.*

Proof. Let $K = K_\infty(\boldsymbol{x})$ with $\boldsymbol{x}$ a system of parameters for R. Also, consider the triangle induced by soft truncations

$$M' \to M \to M'' \to \Sigma M'$$

where $\mathrm{H}^i(M') \cong \mathrm{H}^i(M)$ for $i \leqslant 0$, $(M')^i = 0$ for $i > 0$, and $\mathrm{H}^i(M) \cong \mathrm{H}^i(M'')$ for $i > 0$. As M' and $M' \otimes_R K$ are concentrated in cohomological degrees at most d, it follows that $\mathrm{H}^i(M') = 0$ and $\mathrm{H}^i_\mathfrak{m}(M') = 0$ for all $i > d$. Therefore, for $i > d$, there are the induced horizontal isomorphisms

[2] Proving this, and its converse, was Exercise 1 on Discussion Problems 4

$$\begin{array}{ccc} \mathrm{H}^i_{\mathfrak{m}}(M) & \xrightarrow{\cong} & \mathrm{H}^i_{\mathfrak{m}}(M'') \\ \downarrow & & \downarrow\cong \\ \mathrm{H}^i(M) & \xrightarrow[\cong]{} & \mathrm{H}^i(M'')\,; \end{array}$$

the vertical map on the right is an isomorphism as M'' has $\mathfrak{m}$-power torsion homology and so Theorem 1.4.13. □

Proof of Theorem 4.3.1. Let M, N be MCM R-modules and note that

$$\Omega^2_R(M)^* = \mathrm{RHom}_R(\Omega^2_R(M), R)\,.$$

Now observe that

$$D\,\mathrm{Tr}_R(M) = \mathrm{Hom}_R(\mathrm{Tr}_R(M), E) = \mathrm{Hom}_R(\mathrm{RHom}_R(\Omega^2_R(M), R), E) \simeq \Sigma^d \mathrm{R}\Gamma_{\mathfrak{m}}(\Omega^2(M))\,,$$

where the isomorphism uses local duality; cf. Theorem 1.6.13. This is used in the second isomorphism below:

$$\begin{aligned} D\underline{\mathrm{Hom}}_R(M,N) &\cong \mathrm{Ext}^1_R(N, D\,\mathrm{Tr}_R(M)) \\ &\cong \mathrm{Ext}^1_R(N, \Sigma^d \mathrm{R}\Gamma_{\mathfrak{m}}(\Omega^2_R(M))) \\ &\cong \mathrm{H}^{d+1}(\mathrm{RHom}_R(N, \mathrm{R}\Gamma_{\mathfrak{m}}(\Omega^2_R(M)))) \\ &\cong \mathrm{H}^{d+1}(\mathrm{R}\Gamma_{\mathfrak{m}}\,\mathrm{RHom}_R(N, \Omega^2_R(M))) \\ &\cong \mathrm{H}^{d+1}(\mathrm{RHom}_R(N, \Omega^2(M))) \\ &= \mathrm{Ext}^{d+1}_R(N, \Omega^2(M)) \\ &\cong \underline{\mathrm{Hom}}_R(N, \Omega^{-d-1}(\Omega^2(M))) \\ &\cong \underline{\mathrm{Hom}}_R(N, \Omega^{-d+1}(M))\,; \end{aligned}$$

the first isomorphism is the AR-formula Theorem 4.2.5, the fourth isomorphism uses Proposition 1.4.15, the fifth uses that R is an isolated singularity and Lemma 4.3.3, and the remaining ones are evident. □